# Supplementary Material for "FundusShift: Benchmarking Accuracy and Reliability of Fundus Disease Classification at Unseen Sources, and a Low-Capacity Remedy"

Dang Truong Minh Le

Mohamed bin Zayed University of Artificial Intelligence, Abu Dhabi, United Arab Emirates
dang.le@mbzuai.ac.ae

This supplement gives every table behind the main paper, plus analyses that did not fit there. All tables are generated from the saved results in the repository (`https://doi.org/10.5281/zenodo.23911966`) by `paper/make_assets.py` and `paper/make_supp.py`; numbers are means over seeds unless stated. Abbreviations: bAcc, balanced accuracy; Eq-src, mean of per-source bAcc; R/T/F, raw, temperature scaling (TS), focal temperature scaling (FTS); LP, linear probe; LAC, least-ambiguous conformal sets; G→N, glaucoma images accepted and predicted normal, over all glaucoma images.

# Table of Contents

Supplementary Material for "FundusShift: Benchmarking Accuracy and Reliability of Fundus Disease Classification at Unseen Sources, and a Low-Capacity Remedy" . . . . . 1
*Dang Truong Minh Le*

## S1 Data

### S1.1 Sources, label scope and licences

Table S1 lists every source with its class counts per split. "Scope" records which diseases a source graded: in DR-only sources a normal label means "no DR", and glaucoma was not assessed.

Table S1: Sources. Counts are N/D/G. Development sources contribute train, validation and calibration; test sources and Chákṣu contribute only the evaluation split.

| Source | Country | Scope | Licence | Train | Validation | Calibration | Test or OOD |
|---|---|---|---|---|---|---|---|
| Bangladesh Eye | Bangladesh | full | CC BY 4.0 | 677/1200/925 | 85/150/116 | 84/150/115 | 0/0/0 |
| OIA-DDR | China (147 hospitals, 23 pro | DR only | CC BY-NC-SA 4.0 | 5012/5004/0 | 627/626/0 | 626/625/0 | 0/0/0 |
| DeepDRiD | China | DR only | CC BY-SA 4.0 | 732/868/0 | 94/110/0 | 88/108/0 | 0/0/0 |
| GRAPE | China | full | CC0 | 0/0/470 | 0/0/58 | 0/0/58 | 0/0/0 |
| IDRiD | India | DR only | CC BY 4.0 | 133/277/0 | 17/35/0 | 17/35/0 | 0/0/0 |
| Jichi | Japan | DR only | CC BY 4.0 | 5028/2504/0 | 635/308/0 | 632/308/0 | 0/0/0 |
| SUSTech-SYSU | China | DR only | CC BY 4.0 | 505/470/0 | 63/59/0 | 63/59/0 | 0/0/0 |
| AFIO | Pakistan (Armed Forces Insti | full | CC BY 4.0 | – | – | – | 67/18/44 |
| FIVES | China | full | CC BY 4.0 | – | – | – | 193/200/193 |
| HRF | Germany / Czech Republic | full | CC BY 4.0 | – | – | – | 15/15/15 |
| HYGD | Israel | full | ODC-By 1.0 | – | – | – | 0/0/548 |
| PALM | China (Zhongshan Ophthalmic | full | CC0 | – | – | – | 161/0/0 |
| PAPILA | Spain | full | GPL-3.0 | – | – | – | 333/0/87 |
| Paraguay | Paraguay | DR only | CC BY 4.0 | – | – | – | 708/723/0 |
| Chákṣu | India | full | CC BY 4.0 | – | – | – | 1156/0/187 |

*Reading*. The development side holds 1,395 glaucoma training images, all from Bangladesh Eye (925) and GRAPE (470), whereas DR and normal images come from all seven sources. Six of seven development sources graded only DR. On the test side, only Paraguay is DR-only, HYGD is G-only and PALM is N-only, so per-source bAcc on those two reduces to one recall. These facts explain much of the glaucoma behaviour in the main paper.

### S1.2 Mapping from source labels

Table S2: Native labels mapped to each harmonized class (first token of the stored label; device and vote suffixes omitted).

| Source | Class | Native labels |
|---|---|---|
| Bangladesh Eye | N | Healthy |
| Bangladesh Eye | D | Diabetic_Retinopathy |
| Bangladesh Eye | G | Glaucoma |
| OIA-DDR | N | ddr_0 |
| OIA-DDR | D | ddr_1, ddr_2, ddr_3, ddr_4 |
| DeepDRiD | N | icdr_0 |
| DeepDRiD | D | icdr_1, icdr_2, icdr_3, icdr_4 |
| GRAPE | G | glaucoma_follow_up |
| IDRiD | N | icdr_0 |
| IDRiD | D | icdr_1, icdr_2, icdr_3, icdr_4 |
| Jichi | N | davis_ndr |
| Jichi | D | davis_pdr, davis_ppdr, davis_sdr |
| SUSTech-SYSU | N | icdr_0 |
| SUSTech-SYSU | D | icdr_1, icdr_2, icdr_3, icdr_4, laser_scars(reclassified_icdr_2), laser_scars(reclassified_icdr_3), ... |
| AFIO | N | macula_healthy, od_healthy |
| AFIO | D | macula_ci-dme |
| AFIO | G | od_glaucoma |
| FIVES | N | normal |
| FIVES | D | diabetic_retinopathy |
| FIVES | G | glaucoma |
| HRF | N | healthy |
| HRF | D | diabetic_retinopathy |
| HRF | G | glaucoma |
| HYGD | G | GON+ |
| PALM | N | normal |
| PAPILA | N | healthy |
| PAPILA | G | glaucoma |
| Paraguay | N | 1._No_DR_signs |
| Paraguay | D | 2._Mild_(or_early)_NPDR, 3._Moderate_NPDR, 4._Severe_NPDR, 5._Very_Severe_NPDR, 6._PDR, 7._Advanced_PDR |
| Chákṣu | N | normal |
| Chákṣu | G | glaucoma_suspect |

*Reading.* A native label is mapped only when it supports the class. Grading scales differ: ICDR grades 1–4, Davis SDR–PDR and named DR categories all become D. Glaucoma suspects in Chákṣu follow the source's majority vote.

### S1.3 Leakage control and audit

Duplicates were detected by exact SHA-256 hashes and by a vessel-pattern signature (band-pass green channel, $32 \times 32$) with shift-tolerant normalized cross-correlation at threshold 0.70, then joined with patient identifiers into leakage groups. An automated audit of the released splits found zero violations for each of: a leakage group in more than one split; an identical file in more than one split; a patient in more than one split; a near-duplicate pair across splits; an acquisition family on both development and test sides; a source on both sides; a calibration image without a Cal-A/Cal-B assignment. Brightness-altered copies of GRAPE photographs inside the FIVES glaucoma class were found this way and removed. Of 40,414 inventoried images, 35,280 enter the protocol.

**Excluded collections.** RFMiD 2.0 was used for inventory and duplicate checks only, because class is confounded with camera. A glaucoma collection of optic-disc crops was excluded for its different field of view. OIA-DDR is CC BY-NC-SA 4.0, so derived data remain non-commercial.

**Ethics.** All images are publicly released; this is a secondary analysis. Ethics and consent statements were found in the publications of FIVES, GRAPE, PAPILA, SUSTech-SYSU, DeepDRiD, Jichi, HYGD, Chákṣu, PALM and Paraguay. Statements were not found on the release pages of HRF, Bangladesh Eye and AFIO. Age and sex are available only for PAPILA, so subgroup fairness cannot be assessed.

## S2 Models, Recipes and Prompts

**Table S3.** Encoders. "Fine-tuned" marks the 13 architectures trained end to end; every encoder was also probed frozen.

| Encoder | Pretraining data | Fine-tuned |
|---|---|---|
| deit_b16 | ImageNet-1k | yes |
| convnext_t | ImageNet-1k | yes |
| swin_t | ImageNet-1k | yes |
| inception_v3 | ImageNet-1k | yes |
| efficientnet_b0 | ImageNet-1k | yes |
| densenet121 | ImageNet-1k | yes |
| vit_b16_in21k | ImageNet-21k | yes |
| resnet50 | ImageNet-1k | yes |
| mobilenetv3_l | ImageNet-1k | yes |
| vgg16 | ImageNet-1k | yes |
| vit_b16_mae | ImageNet-1k (MAE, self-supervised) | yes |
| dinov2_b14 | LVD-142M (self-supervised) | no |
| retfound_mae | MEH-MIDAS + public fundus (MAE, RETFound) | yes |
| retfound_dinov2 | MEH-MIDAS fundus (DINOv2, RETFound) | yes |
| clip_b16 | OpenAI web image-text | no |
| metaclip_b16 | MetaCLIP (web image-text) | no |
| siglip_b16 | WebLI | no |
| siglip_l16 | WebLI | no |
| siglip2_b32 | WebLI | no |
| siglip2_b16 | WebLI | no |
| siglip2_l16 | WebLI | no |
| bioclip_b16 | iNat/biodiversity (TreeOfLife-10M) | no |
| bioclip2_l14 | biodiversity (TreeOfLife-200M) | no |
| biomedclip | PMC-15M biomedical figures | no |
| medsiglip | Medical image-text (MedGemma report); fundus: EyePACS | no |
| flair | Retinal image-text; includes five FundusShift test sources | no |

Table S5: Prompt banks used by every zero-shot model and by LANTERN's text branch.

| Bank | Prompts |
|---|---|
| CLIP ensemble | a photo of .; a colour fundus photograph of .; a retinal image of .; a fundus photo showing .; an ophthalmology image of .; a close-up retinal photograph of .; . with class names: N: a normal retina, D: diabetic retinopathy, G: glaucoma |
| CuPL (N) | a healthy fundus with a pink optic disc, a clear macula and regular retinal vessels. a normal retina without haemorrhages, exudates or optic disc abnormality. a fundus photograph with an evenly coloured background and a well-defined foveal reflex. a retina with a small physiological optic cup and an intact neural rim. a fundus free of microaneurysms, cotton-wool spots and neovascularisation. |
| CuPL (D) | a retina with scattered microaneurysms and dot-blot haemorrhages. a fundus with yellow hard exudates near the macula. a retinal photograph with cotton-wool spots and venous beading. a fundus with abnormal new vessels and preretinal haemorrhage. a diabetic eye with intraretinal haemorrhages in several quadrants. |

Table S5: Prompt banks used by every zero-shot model and by LANTERN's text branch.

| Bank | Prompts |
|---|---|
| CuPL (G) | an optic disc with a large cup-to-disc ratio. a fundus with a thinned or notched neural rim of the optic disc. a retina with retinal nerve fibre layer defects next to the optic disc. an optic nerve head with vertical cupping and pallor. a fundus with peripapillary atrophy and glaucomatous optic neuropathy. |
| Single (N) | a colour fundus photograph of a healthy retina with a normal optic disc and no lesions. |
| Single (D) | a colour fundus photograph with diabetic retinopathy showing microaneurysms, haemorrhages and exudates. |
| Single (G) | a colour fundus photograph with glaucoma showing an enlarged optic cup and a thinned neural rim. |

# S3 Complete Benchmark Results

Table S6: Every setting and encoder, with the configuration of highest validation bAcc. †FLAIR's pretraining includes five FundusShift test sources. Multiple-choice models have no validation scores and no probabilities.

| Setting | Encoder | Config | Val | Test | Eq-src | Chákṣu | ECE R/T/F | NLL R/T/F |
|---|---|---|---|---|---|---|---|---|
| 16-shot probe | flair† | none/mbls | .809 | .750 | .684 | .683 | .134/.128/.127 | .947/.915/.891 |
| 16-shot probe | medsiglip | none/ce | .802 | .748 | .683 | .675 | .096/.044/.037 | .701/.644/.644 |
| 16-shot probe | siglip2_l16 | none/mbls | .696 | .621 | .564 | .195 | .124/.075/.074 | .999/.885/.889 |
| 16-shot probe | biomedclip | none/ce | .677 | .479 | .430 | .450 | .198/.113/.120 | 1.18/1.05/1.06 |
| 16-shot probe | dinov2_b14 | none/ce | .675 | .529 | .503 | .192 | .275/.080/.088 | 1.53/.993/.996 |
| 16-shot probe | efficientnet_b0 | none/ce | .665 | .482 | .458 | .279 | .216/.081/.085 | 1.24/1.02/1.03 |
| 16-shot probe | retfound_mae | none/mbls | .662 | .373 | .339 | .207 | .437/.238/.241 | 1.94/1.17/1.17 |
| 16-shot probe | siglip_l16 | none/ce | .658 | .440 | .414 | .316 | .314/.166/.168 | 1.51/1.16/1.16 |
| 16-shot probe | siglip_b16 | none/ce | .653 | .526 | .516 | .266 | .175/.093/.102 | 1.24/1.02/1.03 |
| 16-shot probe | densenet121 | none/ce | .653 | .514 | .499 | .347 | .207/.063/.056 | 1.15/.963/.955 |
| 16-shot probe | convnext_t | none/ce | .649 | .523 | .520 | .403 | .214/.063/.069 | 1.24/.971/.970 |
| 16-shot probe | clip_b16 | none/mbls | .649 | .486 | .419 | .251 | .192/.103/.108 | 1.24/1.08/1.09 |
| 16-shot probe | deit_b16 | none/mbls | .645 | .517 | .484 | .233 | .228/.096/.097 | 1.24/1.01/1.01 |
| 16-shot probe | retfound_dinov2 | none/ce | .644 | .437 | .426 | .158 | .282/.087/.095 | 1.47/1.09/1.09 |
| 16-shot probe | mobilenetv3_l | none/ce | .643 | .450 | .426 | .357 | .277/.106/.113 | 1.48/1.08/1.09 |
| 16-shot probe | siglip2_b16 | none/ce | .638 | .456 | .422 | .281 | .315/.137/.141 | 1.85/1.11/1.11 |
| 16-shot probe | swin_t | none/ce | .634 | .494 | .458 | .204 | .214/.059/.062 | 1.21/1.00/1.01 |
| 16-shot probe | vit_b16_in21k | none/ce | .630 | .495 | .469 | .268 | .234/.087/.090 | 1.34/1.02/1.02 |
| 16-shot probe | resnet50 | none/mbls | .628 | .541 | .509 | .429 | .149/.070/.061 | 1.12/.986/.979 |
| 16-shot probe | inception_v3 | none/mbls | .627 | .457 | .425 | .437 | .323/.099/.101 | 1.59/1.07/1.08 |
| 16-shot probe | siglip2_b32 | none/ce | .622 | .471 | .439 | .375 | .260/.098/.098 | 1.41/1.07/1.07 |
| 16-shot probe | vit_b16_mae | none/ls | .621 | .392 | .412 | .239 | .331/.189/.200 | 1.57/1.20/1.20 |
| 16-shot probe | vgg16 | none/ls | .619 | .468 | .459 | .378 | .213/.135/.125 | 1.73/1.24/1.19 |
| 16-shot probe | metaclip_b16 | none/ce | .613 | .612 | .539 | .458 | .145/.085/.079 | 1.10/.906/.903 |
| 16-shot probe | bioclip2_l14 | none/ce | .612 | .495 | .467 | .407 | .247/.064/.068 | 1.48/1.02/1.03 |
| 16-shot probe | bioclip_b16 | none/ce | .597 | .411 | .386 | .204 | .347/.142/.141 | 1.64/1.14/1.14 |
| CoOp | siglip_b16 | ce | .660 | .491 | .458 | .318 | .236/.175/.177 | 1.35/1.18/1.16 |
| CoOp | biomedclip | ls | .654 | .475 | .437 | .379 | .235/.163/.173 | 1.37/1.19/1.21 |
| CoOp | clip_b16 | ls | .631 | .448 | .382 | .206 | .235/.189/.190 | 1.27/1.16/1.17 |
| CoOp | bioclip2_l14 | mbls | .622 | .474 | .462 | .270 | .235/.133/.134 | 1.50/1.21/1.24 |
| CoOp | siglip_l16 | ce | .612 | .403 | .479 | .368 | .107/.189/.191 | 1.13/1.25/1.28 |
| CoOp | bioclip_b16 | mdca | .593 | .415 | .422 | .152 | .429/.154/.155 | 2.59/1.19/1.20 |
| CoOp | metaclip_b16 | ls | .585 | .540 | .483 | .358 | .125/.089/.084 | .990/.951/.944 |
| CoOp | siglip2_l16 | zs_penalty | .408 | .459 | .489 | .444 | .099/.044/.045 | 1.01/.994/.997 |
| CoOp | siglip2_b32 | zs_penalty | .401 | .399 | .315 | .427 | .045/.191/.186 | 1.05/1.11/1.10 |
| CoOp | siglip2_b16 | zs_penalty | .358 | .335 | .233 | .205 | .132/.224/.224 | 1.10/1.16/1.16 |
| CoOp ctx-init | biomedclip | mdca | .642 | .497 | .447 | .384 | .287/.150/.160 | 1.76/1.19/1.22 |
| CoOp ctx-init | clip_b16 | mdca | .638 | .499 | .429 | .171 | .189/.133/.134 | 1.20/1.08/1.08 |
| CoOp ctx-init | metaclip_b16 | mdca | .634 | .605 | .534 | .418 | .092/.043/.042 | .884/.838/.836 |

Table S6: Every setting and encoder, with the configuration of highest validation bAcc. †FLAIR's pretraining includes five FundusShift test sources. Multiple-choice models have no validation scores and no probabilities.

| Setting | Encoder | Config | Val | Test | Eq-src | Chákṣu | ECE R/T/F | NLL R/T/F |
|---|---|---|---|---|---|---|---|---|
| CoOp ctx-init | bioclip2_l14 | mdca | .611 | .468 | .458 | .231 | .188/.115/.115 | 1.36/1.18/1.20 |
| CoOp ctx-init | bioclip_b16 | mbls | .583 | .444 | .418 | .157 | .419/.173/.175 | 2.56/1.21/1.22 |
| CoOp ctx-init | siglip_l16 | zs_penalty | .442 | .421 | .519 | .480 | .075/.092/.097 | 1.08/1.09/1.09 |
| CoOp ctx-init | siglip_b16 | mdca | .421 | .458 | .513 | .492 | .048/.120/.119 | 1.09/1.06/1.06 |
| CoOp ctx-init | siglip2_l16 | mdca | .386 | .361 | .450 | .499 | .231/.183/.178 | 1.29/1.20/1.23 |
| CoOp ctx-init | siglip2_b32 | zs_penalty | .358 | .380 | .447 | .518 | .054/.061/.058 | 1.06/1.06/1.06 |
| CoOp ctx-init | siglip2_b16 | focal | .351 | .341 | .424 | .496 | .178/.157/.159 | 1.24/1.20/1.22 |
| CoOp + LP | siglip_b16 | mbls | .668 | .496 | .463 | .314 | .243/.176/.179 | 1.37/1.18/1.17 |
| CoOp + LP | biomedclip | mdca | .652 | .482 | .433 | .368 | .292/.177/.186 | 1.68/1.22/1.24 |
| CoOp + LP | siglip_l16 | ce | .647 | .447 | .490 | .327 | .101/.169/.172 | 1.12/1.23/1.27 |
| CoOp + LP | clip_b16 | ls | .637 | .449 | .384 | .213 | .244/.190/.191 | 1.29/1.16/1.16 |
| CoOp + LP | siglip2_l16 | ls | .632 | .559 | .543 | .276 | .154/.030/.030 | .969/.897/.895 |
| CoOp + LP | metaclip_b16 | mbls | .628 | .594 | .506 | .410 | .170/.067/.066 | .959/.860/.857 |
| CoOp + LP | bioclip2_l14 | mbls | .623 | .477 | .463 | .276 | .253/.136/.136 | 1.59/1.21/1.23 |
| CoOp + LP | bioclip_b16 | mdca | .595 | .415 | .424 | .156 | .433/.156/.157 | 2.63/1.19/1.20 |
| CoOp + LP | siglip2_b32 | ce | .539 | .487 | .432 | .459 | .068/.064/.061 | 1.00/.998/.995 |
| CoOp + LP | siglip2_b16 | mbls | .510 | .419 | .369 | .417 | .051/.075/.073 | 1.04/1.04/1.04 |
| CORAL | convnext_t | ce | .883 | .676 | .623 | .349 | .198/.136/.148 | 1.38/1.01/1.11 |
| CORAL | dinov2_lora | ce | .853 | .667 | .625 | .340 | .143/.144/.150 | 1.05/1.05/1.11 |
| ERM | dinov2_lora | ce | .857 | .707 | .646 | .419 | .154/.115/.119 | 1.12/.944/.965 |
| Fine-tune | swin_t | mbls | .893 | .655 | .600 | .298 | .223/.171/.182 | 1.56/1.17/1.31 |
| Fine-tune | convnext_t | ls | .888 | .676 | .611 | .319 | .144/.168/.177 | .926/.982/1.04 |
| Fine-tune | retfound_mae | mdca | .887 | .525 | .417 | .291 | .370/.276/.288 | 2.69/1.50/1.57 |
| Fine-tune | retfound_dinov2 | mbls | .886 | .583 | .501 | .358 | .285/.192/.207 | 1.93/1.19/1.33 |
| Fine-tune | densenet121 | ce | .882 | .626 | .581 | .309 | .238/.139/.158 | 1.88/1.13/1.30 |
| Fine-tune | resnet50 | mdca | .881 | .666 | .620 | .319 | .184/.110/.126 | 1.48/1.05/1.17 |
| Fine-tune | inception_v3 | mbls | .877 | .664 | .624 | .405 | .200/.139/.147 | 1.25/.956/1.05 |
| Fine-tune | vgg16 | mbls | .875 | .671 | .625 | .371 | .169/.135/.147 | 1.28/1.10/1.24 |
| Fine-tune | deit_b16 | mdca | .868 | .622 | .560 | .313 | .230/.158/.167 | 1.88/1.28/1.42 |
| Fine-tune | vit_b16_in21k | mdca | .863 | .596 | .535 | .336 | .251/.180/.190 | 2.02/1.31/1.42 |
| Fine-tune | efficientnet_b0 | ce | .861 | .651 | .596 | .266 | .188/.108/.122 | 1.34/.974/1.04 |
| Fine-tune | vit_b16_mae | ce | .856 | .587 | .532 | .302 | .272/.196/.205 | 2.17/1.42/1.59 |
| Fine-tune | mobilenetv3_l | ls | .846 | .648 | .596 | .238 | .082/.130/.144 | .855/.923/.993 |
| Fine-tune, NLL rule | convnext_t | ce | .870 | .677 | .619 | .324 | .185/.133/.147 | 1.35/1.04/1.16 |
| Fine-tune, NLL rule | resnet50 | ce | .857 | .647 | .601 | .254 | .192/.117/.133 | 1.57/1.12/1.28 |
| Group DRO | convnext_t | ce | .849 | .707 | .649 | .417 | .082/.091/.101 | .886/.926/1.04 |
| Group DRO | dinov2_lora | ce | .798 | .733 | .677 | .486 | .050/.035/.034 | .732/.707/.732 |
| Linear probe | medsiglip | class/ls | .920 | .728 | .654 | .347 | .024/.079/.093 | .647/.694/.739 |
| Linear probe | flair† | class/focal | .900 | .721 | .693 | .703 | .088/.117/.117 | .705/.777/.775 |
| Linear probe | dinov2_b14 | class/mbls | .889 | .674 | .616 | .434 | .104/.103/.115 | .843/.842/.875 |
| Linear probe | retfound_mae | class/ls | .886 | .455 | .351 | .243 | .368/.412/.414 | 1.43/1.71/1.69 |
| Linear probe | retfound_dinov2 | class/ce | .868 | .606 | .522 | .115 | .185/.147/.158 | 1.16/1.03/1.08 |
| Linear probe | siglip2_l16 | class/mbls | .866 | .605 | .562 | .206 | .167/.159/.170 | 1.11/1.08/1.16 |
| Linear probe | vit_b16_mae | class/ls | .866 | .579 | .536 | .102 | .136/.190/.200 | .924/1.02/1.03 |
| Linear probe | vit_b16_in21k | class/ls | .861 | .608 | .572 | .292 | .119/.170/.182 | .983/1.09/1.15 |
| Linear probe | metaclip_b16 | class/ls | .861 | .599 | .551 | .101 | .114/.177/.188 | .908/1.01/1.05 |
| Linear probe | siglip2_b16 | class/ls | .860 | .546 | .520 | .484 | .130/.187/.201 | 1.04/1.15/1.22 |
| Linear probe | siglip_b16 | class/ls | .856 | .567 | .540 | .169 | .130/.189/.201 | .988/1.10/1.13 |
| Linear probe | bioclip2_l14 | class/ls | .853 | .538 | .520 | .365 | .164/.212/.227 | 1.05/1.15/1.17 |
| Linear probe | densenet121 | class/mdca | .852 | .569 | .522 | .121 | .228/.203/.211 | 1.38/1.26/1.35 |
| Linear probe | siglip_l16 | class/ce | .851 | .524 | .508 | .352 | .237/.235/.246 | 1.46/1.45/1.57 |
| Linear probe | swin_t | class/mbls | .850 | .584 | .523 | .176 | .198/.180/.191 | 1.20/1.14/1.17 |
| Linear probe | convnext_t | class/mbls | .848 | .615 | .553 | .218 | .145/.137/.147 | .999/.981/1.01 |
| Linear probe | clip_b16 | class/ls | .844 | .595 | .528 | .477 | .104/.162/.171 | .950/1.07/1.14 |
| Linear probe | deit_b16 | class/mdca | .844 | .595 | .537 | .272 | .179/.155/.167 | 1.11/1.05/1.09 |
| Linear probe | bioclip_b16 | class/mbls | .830 | .539 | .479 | .164 | .207/.198/.212 | 1.22/1.19/1.23 |
| Linear probe | mobilenetv3_l | class/ls | .825 | .583 | .525 | .137 | .089/.153/.165 | .968/1.07/1.11 |

Table S6: Every setting and encoder, with the configuration of highest validation bAcc. †FLAIR's pretraining includes five FundusShift test sources. Multiple-choice models have no validation scores and no probabilities.

| Setting | Encoder | Config | Val | Test | Eq-src | Chákṣu | ECE R/T/F | NLL R/T/F |
|---|---|---|---|---|---|---|---|---|
| Linear probe | siglip2_b32 | class/ls | .824 | .554 | .517 | .461 | .130/.177/.188 | 1.01/1.10/1.13 |
| Linear probe | efficientnet_b0 | class/ce | .824 | .598 | .534 | .196 | .210/.179/.185 | 1.32/1.19/1.18 |
| Linear probe | resnet50 | class/mbls | .815 | .650 | .576 | .334 | .081/.056/.060 | .888/.858/.868 |
| Linear probe | biomedclip | class/mdca | .803 | .479 | .459 | .510 | .274/.271/.281 | 1.34/1.33/1.36 |
| Linear probe | inception_v3 | class/mbls | .799 | .543 | .500 | .347 | .220/.190/.198 | 1.22/1.14/1.15 |
| Linear probe | vgg16 | class/mbls | .793 | .534 | .480 | .267 | .234/.157/.179 | 1.89/1.56/1.36 |
| MLP probe | dinov2_b14 | none/ce | .870 | .598 | .524 | .318 | .263/.172/.183 | 2.31/1.26/1.42 |
| MLP probe | siglip2_l16 | none/ce | .859 | .559 | .481 | .140 | .281/.194/.194 | 2.92/1.73/1.73 |
| MLP probe | metaclip_b16 | none/ce | .853 | .524 | .473 | .122 | .360/.260/.269 | 3.04/1.53/1.69 |
| MLP probe | siglip_b16 | none/ce | .851 | .570 | .542 | .330 | .279/.182/.189 | 2.71/1.42/1.57 |
| MLP probe | siglip_l16 | none/ce | .847 | .546 | .469 | .355 | .346/.212/.216 | 4.72/1.86/2.02 |
| MLP probe | siglip2_b16 | none/ce | .845 | .542 | .510 | .431 | .292/.205/.217 | 2.95/1.71/2.05 |
| MLP probe | bioclip2_l14 | none/ce | .838 | .514 | .457 | .318 | .349/.234/.245 | 3.01/1.53/1.66 |
| MLP probe | clip_b16 | none/ce | .837 | .545 | .487 | .450 | .344/.250/.256 | 4.01/1.94/2.18 |
| MLP probe | bioclip_b16 | none/ce | .827 | .502 | .434 | .089 | .328/.252/.255 | 2.14/1.50/1.56 |
| MLP probe | siglip2_b32 | none/ce | .817 | .546 | .503 | .490 | .302/.181/.189 | 3.26/1.52/1.74 |
| MLP probe | biomedclip | none/ce | .806 | .525 | .506 | .475 | .272/.185/.194 | 2.84/1.85/2.08 |
| SWAD | convnext_t | ce | .883 | .672 | .622 | .309 | .208/.139/.149 | 1.57/1.06/1.16 |
| SWAD | dinov2_lora | ce | .832 | .716 | .655 | .436 | .127/.092/.097 | .964/.854/.879 |
| Tip-Adapter | biomedclip | – | .597 | .478 | .491 | .617 | .332/.091/.096 | 1.75/1.02/1.03 |
| Tip-Adapter | siglip2_l16 | – | .550 | .497 | .534 | .533 | .140/.128/.117 | 1.00/.930/.914 |
| Tip-Adapter | clip_b16 | – | .473 | .456 | .466 | .424 | .082/.071/.077 | 1.04/1.08/1.07 |
| Tip-Adapter (acc. rule) | biomedclip | – | .597 | .478 | .491 | .617 | .332/.091/.096 | 1.75/1.02/1.03 |
| Tip-Adapter (acc. rule) | siglip2_l16 | – | .550 | .497 | .534 | .533 | .140/.128/.117 | 1.00/.930/.914 |
| Tip-Adapter (acc. rule) | clip_b16 | – | .473 | .456 | .466 | .424 | .082/.071/.077 | 1.04/1.08/1.07 |
| Tip-Adapter-F | biomedclip | – | .666 | .495 | .453 | .532 | .225/.147/.152 | 1.28/1.10/1.11 |
| Tip-Adapter-F | siglip2_l16 | – | .602 | .538 | .551 | .528 | .086/.157/.155 | 1.09/.986/.969 |
| Tip-Adapter-F | clip_b16 | – | .564 | .508 | .511 | .459 | .109/.095/.090 | 1.05/.997/.995 |
| Tip-Adapter-F (acc. rule) | biomedclip | – | .666 | .495 | .453 | .532 | .225/.147/.152 | 1.28/1.10/1.11 |
| Tip-Adapter-F (acc. rule) | siglip2_l16 | – | .602 | .538 | .551 | .528 | .086/.157/.155 | 1.09/.986/.969 |
| Tip-Adapter-F (acc. rule) | clip_b16 | – | .564 | .508 | .511 | .459 | .109/.095/.090 | 1.05/.997/.995 |
| WiSE-FT | siglip2_l16 | fully-tuned | .886 | .634 | .587 | .406 | .236/.168/.179 | 1.84/1.24/1.38 |
| Zero-shot | flair† | descriptive_single | .653 | .712 | .624 | .567 | .162/.154/.137 | 1.09/.867/.827 |
| Zero-shot | medsiglip | descriptive_single | .650 | .709 | .606 | .615 | .207/.032/.032 | .838/.690/.690 |
| Zero-shot | biomedclip | descriptive_single | .503 | .471 | .535 | .562 | .229/.061/.074 | 1.29/1.00/1.02 |
| Zero-shot | metaclip_b16 | descriptive_single | .476 | .403 | .406 | .374 | .093/.119/.119 | 1.06/1.07/1.07 |
| Zero-shot | siglip2_b32 | descriptive_single | .473 | .463 | .501 | .505 | .028/.069/.063 | 1.05/1.07/1.07 |
| Zero-shot | siglip2_l16 | clip_ensemble | .461 | .481 | .494 | .374 | .031/.070/.085 | 1.06/1.17/1.13 |
| Zero-shot | siglip_b16 | cupl | .437 | .433 | .508 | .503 | .068/.090/.092 | .959/.975/.973 |
| Zero-shot | siglip2_b16 | descriptive_single | .418 | .509 | .539 | .518 | .107/.176/.167 | 1.06/1.07/1.06 |
| Zero-shot | bioclip_b16 | descriptive_single | .366 | .457 | .518 | .525 | .358/.182/.183 | 3.79/1.10/1.10 |
| Zero-shot | siglip_l16 | cupl | .364 | .357 | .453 | .500 | .280/.162/.165 | 1.24/1.09/1.10 |
| Zero-shot | bioclip2_l14 | clip_ensemble | .344 | .362 | .281 | .074 | .326/.180/.186 | 1.54/1.16/1.18 |
| Zero-shot | clip_b16 | clip_ensemble | .327 | .394 | .445 | .173 | .055/.096/.094 | 1.10/1.15/1.16 |
| Multiple choice | Fundus-R1-7B | avg6 | – | .576 | .569 | .463 | – | – |
| Multiple choice | Qwen2.5-VL-7B-Instruct | avg6 | – | .424 | .462 | .488 | – | – |
| Multiple choice | llava-med-v1.5-mistral-7b-hf | avg6 | – | .347 | .448 | .507 | – | – |
| Multiple choice | llava-onevision-qwen2-7b-ov-hf | avg6 | – | .407 | .327 | .096 | – | – |

*Reading.* Within each setting the encoder ranking on test differs from that on validation (Sect. S5). MedSigLIP is the strongest frozen encoder across probe settings, and FLAIR is comparable only on sources it did not see in pretraining. Fine-tuned RETFound reaches high validation (0.886–0.887) but 0.525–0.583 on test, despite fundus pretraining. TS reduces ECE and NLL for most fine-tuned models but leaves ECE far

**Table S4.** Training recipes. Every setting selects by validation bAcc unless stated; post-hoc calibration always uses Cal-A, thresholds Cal-B.

| | |
|---|---|
| Fine-tuning | Native input size and normalization; AdamW, lr $10^{-4}$ (head $10^{-3}$), wd 0.05, cosine with one warm-up epoch, 15 epochs, batch 32; (source, class)-balanced sampling; crop, flip, rotation and colour augmentation; mixed precision; five losses (CE; label smoothing 0.1; focal $\gamma$=3; MDCA $\beta$=1; MbLS $m$=10, $\lambda$=0.1); 3 seeds. NLL-rule arm: checkpoint by validation NLL. |
| Linear, MLP probes | Frozen embeddings standardized on train; AdamW lr $10^{-3}$, wd $10^{-4}$, 40 epochs, batch 256; weighting none, class or (class, source); MLP: one hidden layer of 512. |
| 16-shot probe | 16 images per class, one per leakage group, redrawn per seed, 300 epochs. |
| Zero-shot | Three prompt banks (Table S5); bank chosen by validation bAcc. |
| CoOp | 16 learnable context tokens, frozen towers, SGD lr 0.002, cosine, 200 full-batch steps on the 16 shots; ctx-init: 4 tokens from "a photo of a"; CoOp+LP: then a linear head, 200 steps, lr $10^{-3}$; loss variants CE, focal, LS, MbLS, MDCA. |
| Tip-Adapter | Official recipe on the same shots; $(\alpha, \beta)$ grid and bank by validation bAcc (official accuracy rule kept as sensitivity); -F: keys fine-tuned 20 epochs, AdamW lr $10^{-3}$. |
| ERM, Group DRO, CORAL | Fine-tuning recipe with ConvNeXt-T, or DINOv2-B/14 with LoRA rank 8 on query and value; Group DRO $\eta$=0.01, group = development source; CORAL $\lambda$=1. |
| SWAD | Dense averaging over segments of 150 updates, official loss-valley rule (3, 6, 0.3); replaces the validation checkpoint. |
| WiSE-FT | Head from the zero-shot classifier, AdamW lr $10^{-5}$, wd 0.1, 15 epochs; interpolation $\alpha \in \{0, 0.1, \ldots, 1\}$ chosen by validation bAcc. |
| Multiple choice | Image plus a three-option question; next-token log-likelihoods of the option letters averaged over all six option orders. |
| LANTERN | Table S36. |

above that of frozen probes. For transparency, the best frozen probe judged by test itself, a class-weighted cross-entropy MedSigLIP linear probe, reached test bAcc 0.778, equal-source 0.702, raw ECE 0.065 and recall G 0.657; validation selected the label-smoothed variant instead.

## S4 Per-Source and Per-Class Results

Table S7: bAcc per unseen source (raw), equal-source mean and worst test source, for LANTERN and the validation-selected representatives.

| Method | Paraguay | HYGD | FIVES | PAPILA | PALM | HRF | AFIO | Chákṣu | Eq-src | Worst |
|---|---|---|---|---|---|---|---|---|---|---|
| LANTERN | .900 | .818 | .735 | .706 | .700 | .622 | .584 | .731 | .724 | .584 |
| 16-shot probe, medsiglip | .874 | .756 | .668 | .499 | .723 | .711 | .548 | .675 | .683 | .499 |
| CoOp, siglip_b16 | .728 | .130 | .489 | .197 | .768 | .415 | .476 | .318 | .458 | .130 |
| CoOp ctx-init, biomedclip | .753 | .038 | .481 | .435 | .563 | .400 | .456 | .384 | .447 | .038 |
| CoOp + LP, siglip_b16 | .735 | .131 | .496 | .197 | .781 | .430 | .471 | .314 | .463 | .131 |
| CORAL, convnext_t | .929 | .263 | .688 | .413 | .911 | .622 | .533 | .349 | .623 | .263 |
| ERM, dinov2_lora | .884 | .505 | .596 | .498 | .816 | .607 | .614 | .419 | .646 | .498 |
| Fine-tune, swin_t | .913 | .237 | .647 | .468 | .836 | .600 | .501 | .298 | .600 | .237 |
| Fine-tune, NLL rule, convnext_t | .917 | .283 | .704 | .449 | .880 | .585 | .516 | .324 | .619 | .283 |
| Group DRO, convnext_t | .920 | .316 | .704 | .592 | .870 | .600 | .539 | .417 | .649 | .316 |
| Linear probe, medsiglip | .934 | .477 | .559 | .493 | .907 | .637 | .571 | .347 | .654 | .477 |
| MLP probe, dinov2_b14 | .889 | .092 | .631 | .285 | .747 | .533 | .492 | .318 | .524 | .092 |
| SWAD, convnext_t | .928 | .243 | .709 | .409 | .901 | .615 | .550 | .309 | .622 | .243 |
| Tip-Adapter, biomedclip | .399 | .744 | .427 | .496 | .480 | .400 | .493 | .617 | .491 | .399 |
| Tip-Adapter-F, biomedclip | .624 | .283 | .465 | .471 | .472 | .407 | .448 | .532 | .453 | .283 |
| WiSE-FT, siglip2_l16 | .881 | .250 | .626 | .423 | .801 | .600 | .529 | .406 | .587 | .250 |
| Zero-shot, medsiglip | .729 | .719 | .723 | .622 | .379 | .556 | .512 | .615 | .606 | .379 |

Table S8: Class recalls on the multi-class test sources.

| Method | FIVES | | | AFIO | | | PAPILA | | Paraguay | |
|---|---|---|---|---|---|---|---|---|---|---|
| | N | D | G | N | D | G | N | G | N | D |
| LANTERN | .687 | .742 | .777 | .323 | .778 | .652 | .819 | .594 | .847 | .953 |
| 16-shot probe, medsiglip | .418 | .757 | .831 | .075 | .796 | .773 | .139 | .858 | .797 | .951 |
| CoOp, siglip_b16 | .477 | .717 | .275 | .164 | 1.00 | .265 | .110 | .284 | .624 | .833 |
| CoOp ctx-init, biomedclip | .522 | .840 | .083 | .373 | .926 | .068 | .349 | .521 | .653 | .852 |
| CoOp + LP, siglip_b16 | .530 | .723 | .233 | .144 | .981 | .288 | .121 | .272 | .625 | .845 |
| CORAL, convnext_t | .440 | .700 | .924 | .687 | .685 | .227 | .405 | .421 | .920 | .937 |
| ERM, dinov2_lora | .171 | .683 | .933 | .443 | .815 | .583 | .413 | .582 | .806 | .963 |
| Fine-tune, swin_t | .328 | .702 | .910 | .483 | .907 | .114 | .318 | .617 | .872 | .954 |
| Fine-tune, NLL rule, convnext_t | .525 | .710 | .877 | .597 | .722 | .227 | .395 | .502 | .896 | .937 |
| Group DRO, convnext_t | .560 | .695 | .858 | .458 | .833 | .326 | .748 | .437 | .917 | .922 |
| Linear probe, medsiglip | .169 | .692 | .817 | .647 | .944 | .121 | .891 | .096 | .917 | .951 |
| MLP probe, dinov2_b14 | .380 | .650 | .864 | .701 | .722 | .053 | .451 | .119 | .879 | .898 |
| SWAD, convnext_t | .520 | .715 | .893 | .657 | .759 | .235 | .362 | .456 | .916 | .940 |
| Tip-Adapter, biomedclip | .245 | .362 | .675 | .353 | .574 | .553 | .015 | .977 | .392 | .406 |
| Tip-Adapter-F, biomedclip | .180 | .670 | .544 | .149 | .870 | .326 | .030 | .912 | .443 | .805 |
| WiSE-FT, siglip2_l16 | .323 | .675 | .881 | .537 | .852 | .197 | .228 | .617 | .834 | .928 |
| Zero-shot, medsiglip | .679 | .800 | .689 | .254 | .556 | .727 | .727 | .517 | .586 | .871 |

Table S9: ECE after TS per source, and the equal-source mean.

| Method | Paraguay | HYGD | FIVES | PAPILA | PALM | HRF | AFIO | Chákṣu | Eq-src |
|---|---|---|---|---|---|---|---|---|---|
| LANTERN | .038 | .069 | .048 | .050 | .109 | .295 | .256 | .035 | .123 |
| 16-shot probe, medsiglip | .075 | .103 | .075 | .321 | .124 | .175 | .301 | .060 | .168 |
| CoOp, siglip_b16 | .106 | .607 | .119 | .446 | .137 | .216 | .388 | .199 | .288 |
| CoOp ctx-init, biomedclip | .090 | .640 | .167 | .242 | .153 | .295 | .356 | .170 | .278 |
| CoOp + LP, siglip_b16 | .103 | .611 | .116 | .443 | .139 | .234 | .412 | .203 | .294 |
| CORAL, convnext_t | .055 | .444 | .184 | .385 | .181 | .291 | .249 | .206 | .256 |
| ERM, dinov2_lora | .044 | .215 | .305 | .288 | .113 | .278 | .234 | .113 | .211 |
| Fine-tune, swin_t | .043 | .480 | .250 | .411 | .170 | .331 | .333 | .239 | .288 |
| Fine-tune, NLL rule, convnext_t | .051 | .423 | .171 | .346 | .175 | .337 | .280 | .211 | .255 |
| Group DRO, convnext_t | .051 | .405 | .156 | .100 | .149 | .263 | .299 | .081 | .203 |
| Linear probe, medsiglip | .032 | .184 | .249 | .063 | .107 | .315 | .285 | .140 | .176 |
| MLP probe, dinov2_b14 | .046 | .617 | .179 | .328 | .130 | .303 | .313 | .215 | .274 |
| SWAD, convnext_t | .051 | .466 | .166 | .390 | .199 | .288 | .242 | .219 | .258 |
| Tip-Adapter, biomedclip | .070 | .315 | .065 | .362 | .184 | .137 | .085 | .170 | .174 |
| Tip-Adapter-F, biomedclip | .053 | .336 | .109 | .568 | .178 | .317 | .315 | .165 | .268 |
| WiSE-FT, siglip2_l16 | .063 | .442 | .250 | .484 | .083 | .338 | .294 | .084 | .279 |
| Zero-shot, medsiglip | .064 | .133 | .052 | .132 | .270 | .192 | .209 | .214 | .150 |

**Table S10.** LANTERN ECE after TS per source on the published split, three seeds pooled, with 95% bootstrap intervals over leakage groups (1,000 draws).

| Source | Images | Groups | ECE (TS) | 95% CI |
|---|---|---|---|---|
| Paraguay | 1,431 | 1,271 | .038 | [.029, .054] |
| HYGD | 548 | 183 | .069 | [.054, .109] |
| FIVES | 586 | 574 | .048 | [.037, .083] |
| PAPILA | 420 | 210 | .050 | [.045, .091] |
| PALM | 161 | 161 | .109 | [.088, .184] |
| HRF | 45 | 45 | .295 | [.185, .432] |
| AFIO | 129 | 65 | .256 | [.196, .365] |
| Chákṣu | 1,343 | 1,306 | .035 | [.032, .061] |

*Reading.* Paraguay is easy for every method, whereas AFIO and HRF are hard for all. D transfers well everywhere; the differences between methods come mostly from G and from N on G-heavy sources. Per-source ECE after a single global temperature remains high for every method on HRF and AFIO, small sets where a pooled ECE hides the error. Table S10 shows that their intervals are wide (HRF 0.185–0.432 with 45 images), so calibration on sets this small cannot be certified either way; a deployment site needs a local calibration set of a few hundred images (Table S14).

## S5 Reliability

Table S11: Selective prediction on test with thresholds fitted on Cal-B, raw/TS/FTS. Cov 80, Cov 90: thresholds for that coverage on Cal-B. Risk 10: SGR bound of 10%.

| | | Cov 80 | | | | Cov 90 | | Risk 10 | | Chákṣu Cov 80 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Method | | Cov. | Risk | G→N | D→N | Cov. | Risk | Cov. | Risk | Cov. | Risk |
| LANTERN | R | .713 | .100 | .059 | .032 | .857 | .139 | .665 | .088 | .354 | .217 |
| | T | .710 | .094 | .056 | .031 | .861 | .139 | .661 | .082 | .365 | .218 |
| | F | .710 | .095 | .058 | .030 | .860 | .138 | .662 | .083 | .372 | .216 |
| 16-shot probe, medsiglip | R | .757 | .196 | .012 | .017 | .883 | .245 | .354 | .060 | .528 | .307 |
| | T | .753 | .195 | .012 | .016 | .876 | .241 | .353 | .057 | .509 | .307 |
| | F | .752 | .195 | .012 | .016 | .876 | .241 | .357 | .058 | .504 | .307 |
| CoOp, siglip_b16 | R | .801 | .477 | .039 | .086 | .898 | .489 | .000 | – | .598 | .648 |
| | T | .797 | .476 | .038 | .089 | .898 | .488 | .000 | – | .572 | .649 |
| | F | .798 | .476 | .038 | .086 | .899 | .489 | .000 | – | .586 | .647 |
| CoOp ctx-init, biomedclip | R | .761 | .454 | .200 | .068 | .869 | .469 | .000 | – | .808 | .395 |
| | T | .745 | .453 | .187 | .065 | .854 | .468 | .000 | – | .811 | .398 |
| | F | .749 | .452 | .189 | .067 | .857 | .468 | .000 | – | .812 | .396 |
| CoOp + LP, siglip_b16 | R | .793 | .470 | .040 | .085 | .895 | .484 | .000 | – | .572 | .673 |
| | T | .796 | .470 | .040 | .090 | .898 | .484 | .000 | – | .551 | .675 |
| | F | .797 | .469 | .039 | .088 | .898 | .484 | .000 | – | .559 | .674 |
| CORAL, convnext_t | R | .668 | .218 | .157 | .029 | .807 | .253 | .727 | .233 | .530 | .354 |
| | T | .655 | .212 | .144 | .028 | .789 | .245 | .706 | .224 | .525 | .353 |
| | F | .668 | .218 | .158 | .029 | .807 | .254 | .726 | .233 | .529 | .354 |
| ERM, dinov2_lora | R | .726 | .228 | .075 | .013 | .848 | .262 | .697 | .224 | .566 | .318 |
| | T | .715 | .223 | .067 | .012 | .829 | .255 | .689 | .221 | .552 | .314 |
| | F | .723 | .227 | .073 | .013 | .839 | .259 | .695 | .222 | .561 | .318 |
| Fine-tune, swin_t | R | .708 | .257 | .156 | .025 | .819 | .283 | .766 | .269 | .457 | .489 |
| | T | .694 | .250 | .147 | .024 | .803 | .276 | .752 | .263 | .444 | .484 |
| | F | .709 | .257 | .156 | .025 | .819 | .283 | .766 | .269 | .458 | .489 |
| Fine-tune, NLL rule, convnext_t | R | .706 | .219 | .141 | .033 | .824 | .253 | .765 | .237 | .497 | .423 |
| | T | .690 | .213 | .126 | .031 | .809 | .246 | .755 | .230 | .489 | .420 |
| | F | .707 | .220 | .143 | .033 | .826 | .254 | .766 | .238 | .497 | .423 |
| Group DRO, convnext_t | R | .734 | .180 | .086 | .042 | .834 | .208 | .661 | .162 | .569 | .249 |
| | T | .738 | .182 | .089 | .042 | .838 | .210 | .664 | .165 | .570 | .249 |
| | F | .751 | .187 | .097 | .044 | .850 | .215 | .668 | .167 | .572 | .249 |
| Linear probe, medsiglip | R | .602 | .127 | .057 | .022 | .750 | .161 | .794 | .176 | .323 | .271 |
| | T | .610 | .129 | .061 | .022 | .765 | .167 | .800 | .178 | .308 | .275 |
| | F | .620 | .132 | .067 | .023 | .771 | .169 | .811 | .183 | .299 | .276 |
| MLP probe, dinov2_b14 | R | .692 | .290 | .304 | .048 | .834 | .335 | .679 | .288 | .535 | .461 |
| | T | .672 | .281 | .281 | .045 | .812 | .326 | .666 | .278 | .524 | .459 |
| | F | .685 | .287 | .298 | .047 | .824 | .330 | .673 | .285 | .532 | .462 |
| SWAD, convnext_t | R | .708 | .225 | .172 | .032 | .832 | .257 | .820 | .253 | .478 | .401 |
| | T | .690 | .216 | .156 | .030 | .813 | .246 | .804 | .245 | .471 | .399 |
| | F | .704 | .223 | .167 | .032 | .825 | .252 | .816 | .250 | .475 | .399 |
| Tip-Adapter, biomedclip | R | .798 | .549 | .095 | .153 | .897 | .554 | .000 | – | .794 | .495 |
| | T | .757 | .547 | .089 | .113 | .874 | .549 | .000 | – | .930 | .490 |
| | F | .762 | .546 | .091 | .116 | .872 | .548 | .000 | – | .923 | .490 |
| Tip-Adapter-F, biomedclip | R | .769 | .526 | .079 | .045 | .882 | .526 | .000 | – | .810 | .409 |
| | T | .771 | .524 | .077 | .046 | .887 | .525 | .028 | .446 | .828 | .416 |
| | F | .770 | .524 | .077 | .046 | .887 | .526 | .030 | .458 | .828 | .416 |
| WiSE-FT, siglip2_l16 | R | .659 | .270 | .100 | .028 | .804 | .306 | .679 | .276 | .430 | .286 |
| | T | .641 | .260 | .088 | .025 | .780 | .295 | .661 | .268 | .417 | .280 |
| | F | .654 | .267 | .097 | .027 | .798 | .303 | .673 | .274 | .426 | .283 |
| Zero-shot, medsiglip | R | .727 | .230 | .122 | .052 | .868 | .270 | .190 | .040 | .745 | .581 |
| | T | .711 | .224 | .107 | .052 | .861 | .267 | .182 | .033 | .689 | .576 |
| | F | .711 | .224 | .107 | .052 | .861 | .267 | .182 | .033 | .689 | .576 |

Table S12: Conformal sets on test, quantiles from Cal-B. Coverage/mean size at nominal 0.90; coverage at nominal 0.95.

| Method | | LAC .90 | APS .90 | Class-LAC .90 | LAC .95 | APS .95 | Class-LAC .95 | Chákṣu LAC .90 |
|---|---|---|---|---|---|---|---|---|
| LANTERN | R | .887/1.20 | .935/1.73 | .882/1.25 | .955 | .954 | .942 | .826/1.42 |
| | T | .885/1.20 | .955/1.89 | .884/1.25 | .958 | .958 | .944 | .828/1.43 |
| | F | .885/1.20 | .922/1.70 | .884/1.25 | .954 | .939 | .941 | .828/1.43 |
| 16-shot probe, medsiglip | R | .913/1.61 | .923/1.99 | .823/1.40 | .966 | .994 | .919 | .930/2.09 |
| | T | .912/1.61 | .922/1.98 | .817/1.39 | .967 | .996 | .916 | .929/2.08 |
| | F | .909/1.60 | .925/2.01 | .822/1.39 | .966 | .996 | .922 | .927/2.09 |
| CoOp, siglip_b16 | R | .761/1.80 | .866/2.24 | .761/1.83 | .843 | .947 | .856 | .817/2.26 |
| | T | .760/1.80 | .862/2.24 | .760/1.83 | .839 | .944 | .858 | .811/2.23 |
| | F | .760/1.79 | .852/2.20 | .773/1.85 | .839 | .986 | .858 | .802/2.18 |
| CoOp ctx-init, biomedclip | R | .777/1.89 | .762/1.91 | .784/1.95 | .846 | .878 | .845 | .854/1.77 |
| | T | .766/1.84 | .760/1.90 | .783/1.94 | .835 | .878 | .846 | .856/1.75 |
| | F | .759/1.81 | .750/1.87 | .782/1.93 | .825 | .875 | .850 | .858/1.75 |
| CoOp + LP, siglip_b16 | R | .755/1.78 | .809/2.01 | .761/1.81 | .839 | .949 | .847 | .802/2.27 |
| | T | .753/1.77 | .808/2.00 | .761/1.81 | .837 | .983 | .849 | .792/2.22 |
| | F | .755/1.77 | .795/1.96 | .773/1.84 | .832 | .985 | .853 | .785/2.18 |
| CORAL, convnext_t | R | .716/1.09 | .815/1.68 | .668/1.13 | .780 | .823 | .754 | .635/1.15 |
| | T | .710/1.07 | .807/1.60 | .663/1.10 | .773 | .819 | .749 | .637/1.15 |
| | F | .713/1.08 | .796/1.56 | .666/1.11 | .769 | .826 | .741 | .637/1.16 |
| ERM, dinov2_lora | R | .728/1.10 | .870/1.72 | .697/1.13 | .801 | .885 | .765 | .684/1.17 |
| | T | .722/1.08 | .870/1.72 | .689/1.11 | .796 | .884 | .758 | .679/1.15 |
| | F | .723/1.08 | .861/1.68 | .692/1.11 | .793 | .882 | .757 | .679/1.15 |
| Fine-tune, swin_t | R | .682/1.08 | .794/1.65 | .635/1.05 | .737 | .803 | .720 | .566/1.18 |
| | T | .676/1.06 | .786/1.59 | .631/1.03 | .732 | .798 | .715 | .559/1.16 |
| | F | .680/1.07 | .784/1.67 | .634/1.04 | .730 | .806 | .714 | .562/1.17 |
| Fine-tune, NLL rule, convnext_t | R | .712/1.08 | .829/1.73 | .679/1.08 | .771 | .838 | .769 | .604/1.18 |
| | T | .708/1.06 | .828/1.71 | .674/1.06 | .764 | .836 | .762 | .604/1.18 |
| | F | .711/1.07 | .806/1.58 | .679/1.07 | .763 | .829 | .757 | .602/1.18 |
| Group DRO, convnext_t | R | .767/1.13 | .844/1.65 | .733/1.09 | .810 | .854 | .814 | .787/1.29 |
| | T | .769/1.13 | .844/1.65 | .733/1.09 | .811 | .854 | .815 | .788/1.29 |
| | F | .771/1.14 | .838/1.65 | .735/1.10 | .811 | .855 | .810 | .790/1.30 |
| Linear probe, medsiglip | R | .743/0.99 | .883/1.68 | .682/0.92 | .857 | .893 | .765 | .631/1.09 |
| | T | .751/1.01 | .888/1.72 | .687/0.94 | .863 | .897 | .765 | .626/1.08 |
| | F | .754/1.01 | .855/1.55 | .691/0.95 | .857 | .876 | .760 | .626/1.08 |
| MLP probe, dinov2_b14 | R | .658/1.11 | .882/2.28 | .637/1.13 | .746 | .931 | .703 | .576/1.17 |
| | T | .651/1.09 | .798/1.79 | .629/1.11 | .739 | .807 | .695 | .573/1.16 |
| | F | .652/1.09 | .777/1.67 | .633/1.12 | .738 | .798 | .696 | .573/1.17 |
| SWAD, convnext_t | R | .698/1.04 | .827/1.73 | .668/1.05 | .761 | .834 | .766 | .562/1.09 |
| | T | .690/1.01 | .822/1.69 | .662/1.02 | .753 | .831 | .758 | .558/1.08 |
| | F | .693/1.02 | .802/1.54 | .666/1.03 | .753 | .825 | .750 | .560/1.09 |
| Tip-Adapter, biomedclip | R | .947/2.50 | .989/2.92 | .785/2.00 | .979 | .989 | .866 | .994/2.27 |
| | T | .970/2.54 | .993/2.95 | .756/1.85 | .988 | .993 | .845 | .000/2.24 |
| | F | .961/2.52 | .992/2.94 | .764/1.90 | .984 | .992 | .840 | .997/2.25 |
| Tip-Adapter-F, biomedclip | R | .819/2.09 | .989/2.92 | .764/1.88 | .889 | .989 | .838 | .897/2.09 |
| | T | .825/2.09 | .992/2.94 | .766/1.86 | .899 | .992 | .841 | .901/2.10 |
| | F | .817/2.08 | .948/2.70 | .763/1.87 | .889 | .990 | .837 | .898/2.10 |
| WiSE-FT, siglip2_l16 | R | .676/1.12 | .805/1.78 | .633/1.13 | .738 | .813 | .707 | .717/1.23 |
| | T | .668/1.10 | .804/1.77 | .628/1.11 | .732 | .812 | .700 | .707/1.20 |
| | F | .671/1.11 | .790/1.70 | .631/1.12 | .729 | .811 | .701 | .709/1.21 |
| Zero-shot, medsiglip | R | .923/1.74 | .896/1.80 | .912/1.87 | .981 | .996 | .967 | .852/1.91 |
| | T | .924/1.76 | .920/1.89 | .916/1.90 | .982 | .994 | .968 | .842/1.90 |
| | F | .924/1.76 | .920/1.89 | .916/1.90 | .982 | .994 | .968 | .842/1.90 |

*Reading.* Methods whose sets stay near nominal coverage (16-shot, zero-shot) do so with mean set sizes of 1.6 to 1.8 out of three classes, which is of little use. Methods with compact sets lose coverage. TS keeps the ranking of confidence, so it barely changes coverage or risk at a fixed threshold.

*Reading.* On the other half of Cal-B, the promises hold almost exactly; on test and Chákṣu they fail. The failure therefore comes from the new sources, not from a small calibration set.

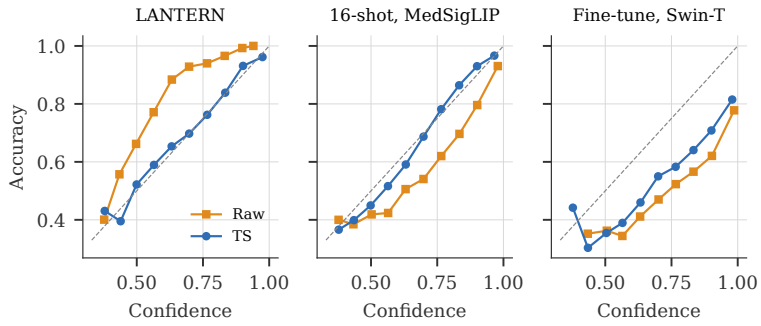



**Fig. S1.** Test reliability on the published split, seeds pooled, bins with at least 30 images. Raw LANTERN is underconfident and TS brings it near the diagonal; fine-tuned Swin-T stays overconfident after TS.

Table S14: Calibration budget: fitted $T$ on all of Cal-A, the oracle $T$ fitted on test, and test NLL after TS with natural random subsets of Cal-A (median of 100 draws).

| Setting | Model | $T$ | $T$ oracle | NLL, 50 | NLL, 200 | NLL, 1000 | NLL, all |
|---|---|---|---|---|---|---|---|
| 16-shot probe | biomedclip | 1.59 | 2.88 | 1.05 | 1.05 | 1.05 | 1.05 |
| 16-shot probe | clip_b16 | 1.80 | 3.28 | 1.08 | 1.08 | 1.07 | 1.07 |
| 16-shot probe | dinov2_b14 | 2.80 | 4.47 | .997 | .994 | .993 | .993 |
| 16-shot probe | resnet50 | 3.75 | 2.61 | .979 | .976 | .977 | .977 |
| 16-shot probe | siglip2_l16 | 1.57 | 1.75 | .929 | .926 | .926 | .926 |
| CoOp | biomedclip | 2.07 | 6.48 | 1.23 | 1.22 | 1.23 | 1.23 |
| CoOp | clip_b16 | 1.43 | 4.17 | 1.21 | 1.20 | 1.19 | 1.20 |
| CoOp | siglip2_l16 | .780 | .893 | 1.07 | 1.07 | 1.07 | 1.07 |
| CoOp ctx-init | biomedclip | 2.22 | 6.34 | 1.19 | 1.17 | 1.18 | 1.18 |
| CoOp ctx-init | clip_b16 | 1.25 | 3.88 | 1.20 | 1.19 | 1.19 | 1.19 |
| CoOp ctx-init | siglip2_l16 | 1.32 | 3.97 | 1.20 | 1.20 | 1.19 | 1.19 |
| CoOp + LP | biomedclip | 2.17 | 6.89 | 1.24 | 1.23 | 1.24 | 1.24 |
| CoOp + LP | clip_b16 | 1.51 | 4.39 | 1.21 | 1.20 | 1.19 | 1.19 |
| CoOp + LP | siglip2_l16 | .358 | .385 | .898 | .896 | .896 | .895 |
| Fine-tune | resnet50 | 1.52 | 3.65 | 1.02 | 1.01 | 1.01 | 1.01 |
| Fine-tune, NLL rule | resnet50 | 1.69 | 4.68 | 1.13 | 1.11 | 1.12 | 1.12 |
| Linear probe | biomedclip | .866 | 1.79 | 1.06 | 1.04 | 1.04 | 1.04 |
| Linear probe | clip_b16 | .889 | 1.83 | .971 | .944 | .951 | .950 |
| Linear probe | dinov2_b14 | .999 | 2.04 | .926 | .891 | .904 | .901 |
| Linear probe | resnet50 | 1.16 | 1.87 | .924 | .911 | .911 | .910 |
| Linear probe | siglip2_l16 | .999 | 2.22 | 1.01 | .989 | .993 | .993 |
| Multiple choice | Fundus-R1-7B | 1.87 | 2.24 | .821 | .816 | .817 | .817 |
| Multiple choice | Qwen2.5-VL-7B-Instruct | 1.78 | 3.39 | 1.07 | 1.08 | 1.08 | 1.08 |
| Multiple choice | llava-med-v1.5-mistral-7b-hf | .725 | .483 | 1.07 | 1.07 | 1.07 | 1.07 |
| Multiple choice | llava-onevision-qwen2-7b-ov-hf | 1.96 | 11.32 | 1.30 | 1.28 | 1.28 | 1.28 |
| MLP probe | biomedclip | 1.74 | 13.47 | 1.90 | 1.85 | 1.85 | 1.85 |
| MLP probe | clip_b16 | 2.36 | 16.03 | 1.99 | 1.97 | 1.95 | 1.94 |
| MLP probe | dinov2_b14 | 2.09 | 7.61 | 1.32 | 1.25 | 1.26 | 1.26 |
| MLP probe | siglip2_l16 | 1.91 | 11.20 | 1.78 | 1.75 | 1.74 | 1.73 |
| Zero-shot | biomedclip | 1000.00 | 137.44 | 1.10 | 1.10 | 1.10 | 1.10 |
| Zero-shot | clip_b16 | .785 | 2.12 | 1.15 | 1.15 | 1.15 | 1.15 |
| Zero-shot | siglip2_l16 | .604 | 1.69 | 1.17 | 1.18 | 1.17 | 1.17 |

*Reading.* About 200 calibration images give nearly the NLL of the full set, yet the fitted $T$ stays far from the test oracle for several models: more calibration data from the same sources does not close the shift.

**Table S13.** Is the failure caused by too few calibration images? Cal-B is split by leakage group 100 times; thresholds fit on one half are read on the other half (same sources), on test and on Chákṣu (TS).

| | Coverage at Cov 80 | | | Risk at Risk 10 | | | LAC .90 coverage | | |
|---|---|---|---|---|---|---|---|---|---|
| Method | Other half | Test | Chákṣu | Other half | Test | Chákṣu | Other half | Test | Chákṣu |
| 16-shot probe, medsiglip | .799 | .754 | .507 | .054 | .041 | .132 | .900 | .911 | .926 |
| CoOp, siglip_b16 | .799 | .797 | .573 | – | – | – | .901 | .760 | .811 |
| CoOp ctx-init, biomedclip | .800 | .744 | .810 | .134 | .257 | .294 | .900 | .767 | .856 |
| CoOp + LP, siglip_b16 | .799 | .795 | .548 | – | – | – | .901 | .756 | .796 |
| CORAL, convnext_t | .797 | .650 | .522 | .068 | .209 | .357 | .901 | .710 | .632 |
| ERM, dinov2_lora | .799 | .714 | .550 | .068 | .210 | .289 | .900 | .722 | .682 |
| Fine-tune, swin_t | .800 | .695 | .446 | .069 | .251 | .487 | .901 | .675 | .554 |
| Fine-tune, NLL rule, convnext_t | .798 | .688 | .487 | .068 | .213 | .424 | .902 | .709 | .605 |
| Group DRO, convnext_t | .797 | .737 | .569 | .064 | .150 | .248 | .900 | .771 | .789 |
| Linear probe, medsiglip | .800 | .610 | .308 | .070 | .161 | .318 | .901 | .752 | .632 |
| MLP probe, dinov2_b14 | .801 | .673 | .526 | .069 | .258 | .452 | .901 | .651 | .573 |
| SWAD, convnext_t | .798 | .689 | .470 | .070 | .230 | .423 | .901 | .691 | .560 |
| Tip-Adapter, biomedclip | .797 | .757 | .928 | – | – | – | .900 | .970 | .000 |
| Tip-Adapter-F, biomedclip | .801 | .772 | .829 | .084 | .436 | .642 | .900 | .827 | .904 |
| WiSE-FT, siglip2_l16 | .799 | .641 | .420 | .068 | .258 | .281 | .900 | .668 | .708 |
| Zero-shot, medsiglip | .799 | .717 | .696 | .045 | .027 | .432 | .902 | .925 | .844 |

**Table S15.** Trivial predictors, the floor for every metric.

| View | Predictor | Acc. | bAcc | Eq-src | ECE | NLL |
|---|---|---|---|---|---|---|
| Test | uniform | .445 | .333 | .429 | .112 | 1.10 |
| Test | train_prior | .445 | .333 | .429 | .063 | 1.30 |
| Test | always_N | .445 | .333 | .429 | – | – |
| Test | always_D | .288 | .333 | .214 | – | – |
| Test | always_G | .267 | .333 | .357 | – | – |
| Chákṣu | uniform | .861 | .500 | .500 | .527 | 1.10 |
| Chákṣu | train_prior | .861 | .500 | .500 | .353 | .978 |
| Chákṣu | always_N | .861 | .500 | .500 | – | – |
| Chákṣu | always_D | .000 | .000 | .000 | – | – |
| Chákṣu | always_G | .139 | .500 | .500 | – | – |

*Reading.* A class-prior predictor reaches ECE 0.063 with chance bAcc, so low ECE alone says nothing about usefulness. The high overall rank correlation (0.781 here, over 847 rows that also count the accuracy-rule Tip-Adapter variants; 0.786 over the 816 configurations of the main paper) mostly reflects differences between families; within fine-tuning it is 0.053.

## S6 Robustness Checks

*Reading.* Validation exceeded test in 98% of probe rows, risk exceeded its internal estimate in 89%, and LAC fell below its internal coverage in 84%; for LANTERN the shares were 94%, 88% and 94%. Ranking is not stable either: the 16-shot probe, best on the published split among representatives, is below full training on every encoder over the rotations.

**Table S16.** Spearman correlation between validation and test bAcc across configurations within each setting.

| Setting | Configurations | Spearman |
|---|---|---|
| 16-shot probe | 130 | .374 |
| CoOp | 70 | .800 |
| CoOp ctx-init | 70 | .927 |
| CoOp + LP | 50 | .393 |
| Fine-tune | 65 | .053 |
| Linear probe | 390 | .344 |
| MLP probe | 11 | .527 |
| Tip-Adapter | 3 | .500 |
| Tip-Adapter (acc. rule) | 3 | .500 |
| Tip-Adapter-F | 3 | −.500 |
| Tip-Adapter-F (acc. rule) | 3 | −.500 |
| WiSE-FT | 4 | -1.00 |
| Zero-shot | 36 | .831 |
| All configurations | 847 | .781 |

**Table S17.** Fifty random 7/7 source rotations (published split excluded). Probes use class-weighted cross-entropy; arrows go from internal estimates on held-back Cal-B to unseen test sources, after TS.

| Method | Val | Test | 5–95% | Eq-src | Rec. G | ECE raw | ECE TS | Risk at Cov 80 | LAC .90 | Coverage .80 |
|---|---|---|---|---|---|---|---|---|---|---|
| LANTERN | .853 | .771 | .706–.827 | .747 | .667 | .146 | .033 | .080→.116 | .898→.853 | .796→.742 |
| medsiglip, full | .922 | .745 | .621–.827 | .725 | .637 | .095 | .084 | .042→.134 | .901→.769 | .795→.668 |
| medsiglip, 16-shot | .814 | .716 | .626–.817 | .713 | .723 | .123 | .056 | .164→.217 | .899→.868 | .803→.778 |
| dinov2_b14, full | .892 | .655 | .529–.765 | .650 | .531 | .123 | .118 | .077→.221 | .899→.735 | .794→.659 |
| dinov2_b14, 16-shot | .713 | .552 | .432–.659 | .571 | .505 | .192 | .073 | .296→.383 | .900→.860 | .801→.779 |
| siglip2_l16, full | .876 | .631 | .549–.731 | .617 | .549 | .155 | .141 | .102→.280 | .900→.728 | .804→.680 |
| siglip2_l16, 16-shot | .722 | .578 | .476–.683 | .562 | .586 | .160 | .080 | .292→.372 | .901→.871 | .796→.751 |
| convnext_t, full | .860 | .593 | .483–.700 | .609 | .426 | .149 | .130 | .112→.287 | .901→.730 | .800→.684 |
| convnext_t, 16-shot | .679 | .494 | .379–.604 | .504 | .456 | .223 | .091 | .352→.463 | .900→.840 | .795→.771 |

*Reading.* Text fusion is the component that matters most, in both accuracy and calibration. The bag is neutral, and between-site projection lowers accuracy across rotations although it helped on development data (Sect. S8).

*Reading.* Seed variation is an order of magnitude smaller than variation between source partitions, so single-seed rotations measure the partition effect. The projection-free variant stays ahead with seeds averaged.

*Reading.* The between-site directions hold 79–87% of every class-mean difference, about four times what a random subspace holds. Sites differ in severity and case mix within each label, so their within-class differences are themselves disease directions; projecting them out removes signal, which explains the accuracy loss over the rotations.

*Reading.* The source is almost perfectly decodable from every representation, frozen or fine-tuned (0.93–1.00), even within one class. Fine-tuning does not add site information, so its failure lies in what its classifier relies on, not in what its features contain; frozen features carry the same site signal, and a low-capacity head simply exploits it less. The between-site projection removes site information for the development sources (0.962 to

**Table S18.** LANTERN ablation over the 50 rotations (TS); standard deviation of bAcc across rotations.

| Configuration | bAcc | SD | Eq-src | Rec. N | Rec. D | Rec. G | False G | ECE | Chákṣu |
|---|---|---|---|---|---|---|---|---|---|
| LANTERN | .771 | .040 | .747 | .796 | .850 | .667 | .081 | .033 | .666 |
| without projection | .788 | .041 | .766 | .813 | .845 | .708 | .071 | .034 | .692 |
| without bag | .768 | .040 | .745 | .795 | .852 | .657 | .080 | .033 | .663 |
| without text | .754 | .047 | .728 | .776 | .827 | .658 | .100 | .056 | .626 |
| single head only | .770 | .055 | .740 | .792 | .818 | .701 | .088 | .053 | .593 |
| text only | .713 | .036 | .692 | .693 | .831 | .614 | .131 | .022 | .594 |

**Table S19.** Seed variability over the rotations: three training seeds per rotation (rotations where all three finished). Seed SD: standard deviation across seeds within a rotation, averaged; rotation SD: standard deviation of seed-averaged bAcc across rotations.

| Configuration | Rotations | bAcc | Seed SD | Rotation SD | ECE | Recall G |
|---|---|---|---|---|---|---|
| LANTERN | 50 | .771 | .003 | .039 | .033 | .666 |
| without projection | 50 | .789 | .004 | .039 | .034 | .709 |
| text only | 50 | .713 | .000 | .036 | .022 | .614 |
| Without projection minus LANTERN, seed-averaged: +0.018, higher on 45 of 50 rotations | | | | | | |

0.354) but not for unseen ones (0.998), while it costs disease signal (0.890 to 0.855): it removes the known sites' directions, and new sites differ along new ones.

*Reading.* Glaucoma recall grows with the number of glaucoma sources on the development side, so the weakness is one of site diversity, which more sources can address. LANTERN's advantage over the linear probe is largest when diversity is lowest (two sources: bAcc 0.761 against 0.678).

*Reading.* N–D and D–G stay near 0.90–0.95 while N–G is 0.64–0.75, so glaucoma is a ranking failure, not a threshold failure. Within-source learning recovers much of it (Chákṣu gap +0.372), so the labels are learnable and the loss is tied to the shift. AFIO stays low even within source.

*Reading.* More sources usually help on test (MedSigLIP 0.594 to 0.817 from one to seven sources) but not on Chákṣu (0.534 to 0.350), and more labels from the same sources give little after about 64 per class. Shift is not solved by more of the same data.

## S7 Attempted Fixes and Other Probes

*Reading.* No label-free fix helps reliably. TENT changes little, BatchNorm adaptation harms ResNet-50 (notably on single-class PALM), and T3A is mostly negative. EM prior estimation is unstable and usually worsens ECE. True per-source priors raise bAcc to 0.81–0.86 for most settings, so class balance is a large part of the shift, but this oracle uses test labels, and single-class sources make it easy.

**Table S20.** Why projection hurts. Share of the norm of each class-mean difference that lies in the removed between-site subspace ($k$ directions), on the development data used by each head, against a random subspace of the same size; and alignment between the two glaucoma sites' difference and the N–G direction.

| Source left out | $k$ | N–D | N–G | D–G | N–G, random | \| cos \| G-site, N–G |
|---|---|---|---|---|---|---|
| none (all) | 14 | .868 | .788 | .811 | .187 | .098 |
| Bangladesh Eye | 11 | .839 | .548 | .602 | .225 | – |
| OIA-DDR | 12 | .830 | .839 | .817 | .208 | .073 |
| DeepDRiD | 12 | .850 | .776 | .762 | .208 | .083 |
| GRAPE | 13 | .866 | .819 | .836 | .147 | – |
| IDRiD | 12 | .818 | .784 | .793 | .209 | .100 |
| Jichi | 12 | .832 | .726 | .747 | .218 | .048 |
| SUSTech-SYSU | 12 | .791 | .785 | .782 | .207 | .115 |

**Table S21.** How much of the source can be read from each representation. Five-fold cross-validated balanced accuracy of a logistic probe (C = 0.05) predicting the disease, the source, or the source within normal or DR images only, on development (validation and calibration) rows and on test rows. Chance for the source is 1/6 or 1/7.

| Features | Rows | $n$ | Disease | Source | Source within N | Source within D |
|---|---|---|---|---|---|---|
| Fine-tuned Swin-T | development | 5,951 | .891 | .951 | .963 | .931 |
| Fine-tuned Swin-T | test | 3,320 | .908 | .977 | .996 | .996 |
| Group DRO ConvNeXt-T | development | 5,951 | .867 | .930 | .933 | .909 |
| Group DRO ConvNeXt-T | test | 3,320 | .906 | .980 | .994 | .944 |
| ERM DINOv2 LoRA | development | 5,951 | .873 | .946 | .967 | .914 |
| ERM DINOv2 LoRA | test | 3,320 | .911 | .986 | .998 | .995 |
| Frozen MedSigLIP (pooled) | development | 5,951 | .895 | .961 | .949 | .949 |
| Frozen MedSigLIP (pooled) | test | 3,320 | .931 | .998 | .999 | .986 |
| Frozen MedSigLIP tokens (LANTERN input) | development | 5,951 | .890 | .962 | .958 | .959 |
| Frozen MedSigLIP tokens (LANTERN input) | test | 3,320 | .924 | .998 | .999 | .986 |
| LANTERN input after projection | development | 5,951 | .855 | .354 | .318 | .322 |
| LANTERN input after projection | test | 3,320 | .924 | .998 | .999 | .984 |

Table S29: $K$ labelled images from each target source, read on the rest of that source: ECE with the frozen $T$ against $T$ re-fitted on $K$; LAC .90 coverage with the frozen quantile against a quantile from $K$, with the 5th percentile over draws.

| Setting | Model | $K$ | ECE frozen | ECE $T$ from $K$ | LAC frozen | LAC from $K$ | LAC 5% |
|---|---|---|---|---|---|---|---|
| 16-shot probe | medsiglip | 10 | .156 | .128 | .879 | .897 | .714 |
| | | 25 | .152 | .093 | .869 | .920 | .801 |
| | | 50 | .154 | .087 | .869 | .899 | .808 |
| | | 100 | .129 | .067 | .900 | .901 | .837 |
| Linear probe | medsiglip | 10 | .172 | .134 | .677 | .899 | .704 |
| | | 25 | .153 | .098 | .682 | .917 | .785 |
| | | 50 | .154 | .086 | .682 | .898 | .797 |
| | | 100 | .134 | .073 | .672 | .901 | .841 |
| Zero-shot | medsiglip | 10 | .163 | .136 | .884 | .901 | .722 |
| | | 25 | .155 | .103 | .884 | .920 | .809 |
| | | 50 | .156 | .094 | .884 | .900 | .813 |
| | | 100 | .121 | .080 | .919 | .902 | .843 |
| ERM | dinov2_lora | 10 | .200 | .140 | .658 | .904 | .714 |
| | | 25 | .189 | .110 | .663 | .920 | .819 |
| | | 50 | .191 | .104 | .663 | .899 | .805 |
| | | 100 | .194 | .089 | .643 | .901 | .831 |
| Group DRO | convnext_t | 10 | .189 | .140 | .712 | .903 | .715 |
| | | 25 | .178 | .117 | .721 | .916 | .803 |
| | | 50 | .180 | .109 | .721 | .899 | .800 |
| | | 100 | .160 | .101 | .718 | .901 | .837 |

Table S29: $K$ labelled images from each target source, read on the rest of that source: ECE with the frozen $T$ against $T$ re-fitted on $K$; LAC .90 coverage with the frozen quantile against a quantile from $K$, with the 5th percentile over draws.

| Setting | Model | $K$ | ECE frozen | ECE $T$ from $K$ | LAC frozen | LAC from $K$ | LAC 5% |
|---|---|---|---|---|---|---|---|
| Fine-tune | swin_t | 10 | .283 | .125 | .590 | .904 | .712 |
| | | 25 | .276 | .095 | .588 | .921 | .809 |
| | | 50 | .277 | .088 | .588 | .901 | .811 |
| | | 100 | .285 | .066 | .557 | .902 | .838 |

*Reading.* Ten labels from the new source already bring LAC coverage near 0.90 on average and re-fitted $T$ lowers ECE, but the worst draws stay far below nominal at small $K$; this is a deployment recipe, not a guarantee.

*Reading.* CLAHE does not help fine-tuning or the 16-shot probe. The MedSigLIP linear probe rises on test, but its validation falls, so validation-based selection would not adopt it.

*Reading.* MSP accepts a third to two thirds of out-of-scope images. KNN rejects most of them and lowers risk, but largely by dropping whole hard sources; it is useful for flagging strange images rather than as a remedy for shift.

*Reading.* Generative multimodal models rarely call glaucoma. Mild DR (grade 1) is the hardest D grade for most models. Strong blur and compression mainly remove glaucoma recall, consistent with the disc being a small, detail-dependent region.

## S8 LANTERN: Specification and Development

### S8.1 Design search on development sources

All results in this section are leave-one-development-source-out on development data, equal-site mean, three seeds unless noted, and use no test row. Table S37 compares losses (cross-entropy; a conditional-logit pair loss across sites, "cross"; the same within sites, "strata"), a single head against a leave-one-source-out bag, the between-site projection ("nap"), and text fusion with one bank or all three, on pooled MedSigLIP features.

Table S37: Factorial comparison on pooled features (development, three seeds).

| Arm | bAcc | ECE | NLL | Recall G |
|---|---|---|---|---|
| cross bag nap+allbanks | .794 | .156 | .574 | .636 |
| strata bag nap+allbanks | .792 | .151 | .573 | .625 |
| ce bag nap+allbanks | .791 | .155 | .581 | .639 |
| strata bag+allbanks | .784 | .148 | .584 | .618 |
| cross bag nap+text | .784 | .137 | .577 | .633 |
| cross bag+allbanks | .782 | .149 | .594 | .557 |
| strata single nap+allbanks | .782 | .144 | .578 | .692 |
| strata bag nap+text | .781 | .131 | .577 | .630 |
| ce bag nap+text | .780 | .134 | .584 | .640 |
| ce bag+allbanks | .777 | .149 | .604 | .576 |
| cross single nap+allbanks | .776 | .141 | .589 | .574 |
| strata bag+text | .774 | .129 | .589 | .622 |
| strata single nap+text | .773 | .128 | .583 | .696 |
| cross bag+text | .773 | .134 | .597 | .570 |
| cross single nap+text | .770 | .125 | .593 | .589 |
| strata single+allbanks | .770 | .135 | .595 | .594 |

Table S37: Factorial comparison on pooled features (development, three seeds).

| Arm | bAcc | ECE | NLL | Recall G |
|---|---|---|---|---|
| ce bag+text | .768 | .132 | .606 | .589 |
| ce single nap+allbanks | .765 | .136 | .601 | .583 |
| strata single+text | .762 | .120 | .599 | .607 |
| cross bag nap | .762 | .133 | .623 | .625 |
| cross single+allbanks | .760 | .135 | .616 | .472 |
| strata bag nap | .759 | .127 | .621 | .603 |
| ce single nap+text | .758 | .117 | .604 | .585 |
| cross single+text | .753 | .123 | .619 | .485 |
| ce single+allbanks | .752 | .143 | .622 | .531 |
| ce bag nap | .750 | .129 | .641 | .629 |
| strata bag | .747 | .130 | .640 | .597 |
| ce single+text | .746 | .130 | .624 | .541 |
| cross bag | .737 | .146 | .663 | .481 |
| strata single nap | .731 | .107 | .651 | .712 |
| cross single nap | .729 | .126 | .666 | .526 |
| strata single | .723 | .129 | .675 | .565 |
| ce bag | .721 | .145 | .687 | .507 |
| ce single nap | .704 | .125 | .694 | .517 |
| cross single | .691 | .158 | .715 | .319 |
| ce single | .675 | .169 | .741 | .417 |

*Reading*. Every component adds a little alone; with all three in place the loss choice no longer matters (0.791–0.794). Text with all banks is the largest single step.

Table S38: Head input (pooled feature, last-layer token mean, three depths, both) and head recipe (base: 40 epochs, lr $10^{-3}$, wd $10^{-4}$, batch 256; reg: 12 epochs, lr $3 \times 10^{-4}$, wd $10^{-2}$, batch 64), single head or bag with projection, with text.

| Study | Arm | bAcc | ECE | NLL | Recall G | False G |
|---|---|---|---|---|---|---|
| recipe | last reg bag nap+text | .792 | .151 | .575 | .764 | .100 |
| recipe | pooled base bag nap+text | .791 | .155 | .581 | .639 | .070 |
| input | pooled bag nap+text | .791 | .155 | .581 | .639 | .070 |
| recipe | last base bag nap+text | .787 | .150 | .582 | .678 | .078 |
| input | last bag nap+text | .787 | .150 | .582 | .678 | .078 |
| recipe | pooled reg bag nap+text | .787 | .155 | .580 | .783 | .109 |
| input | pooled last bag nap+text | .782 | .154 | .594 | .707 | .093 |
| recipe | pooled reg single+text | .759 | .136 | .621 | .688 | .109 |
| recipe | last reg single+text | .759 | .133 | .616 | .650 | .098 |
| recipe | last base single+text | .757 | .132 | .616 | .547 | .074 |
| input | last single+text | .757 | .132 | .616 | .547 | .074 |
| recipe | pooled base single+text | .752 | .143 | .622 | .531 | .071 |
| input | pooled single+text | .752 | .143 | .622 | .531 | .071 |
| input | pooled last single+text | .744 | .149 | .631 | .546 | .085 |
| input | depths bag nap+text | .731 | .118 | .642 | .613 | .118 |
| input | depths single+text | .696 | .139 | .678 | .520 | .126 |

*Reading*. Heavier or learned modules repeatedly lost the glaucoma class at held-out sites, which led to the low-capacity design.

## S8.2 Error analysis and disc-side bias

*Reading*. On OIA-DDR, 1,629 of 5,639 normal images were predicted G although the source holds no glaucoma. Between 0.9% and 13.7% of a site's images were wrong

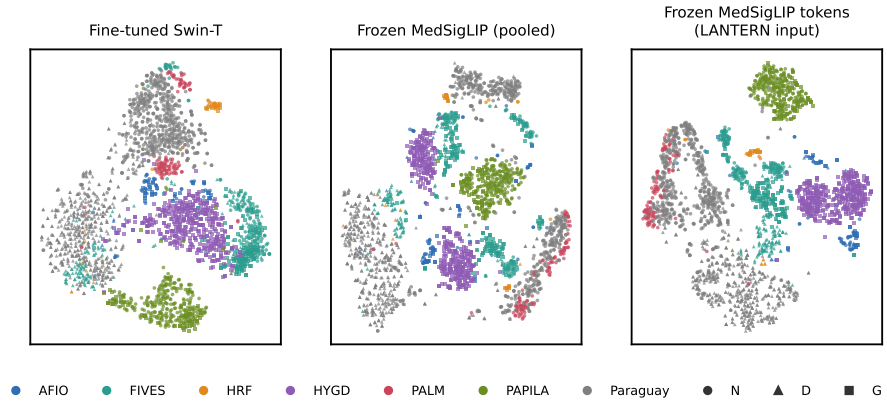



**Fig. S2.** t-SNE of test images (2,500 sampled), coloured by source, marker by class, for a fine-tuned encoder and the two frozen MedSigLIP representations. Every representation clusters by source.

**Table S22.** Glaucoma recall over the 50 rotations by the number of development sources that contain glaucoma.

| Method | G sources in development | Rotations | Recall G | Test bAcc |
|---|---|---|---|---|
| LANTERN | 2 | 8 | .554 | .761 |
| LANTERN | 3 | 18 | .672 | .773 |
| LANTERN | 4 | 17 | .700 | .773 |
| LANTERN | 5 | 6 | .690 | .764 |
| LANTERN | 6 | 1 | .762 | .806 |
| MedSigLIP LP | 2 | 8 | .469 | .678 |
| MedSigLIP LP | 3 | 18 | .628 | .739 |
| MedSigLIP LP | 4 | 17 | .667 | .763 |
| MedSigLIP LP | 5 | 6 | .747 | .773 |
| MedSigLIP LP | 6 | 1 | .959 | .889 |

under all nine inputs; among normal IDRiD images, 28%. A global class offset did not fix the G confusion, so it is a ranking problem.

Table S43: Upgrades of LANTERN's input on development data: dimension (384 or 1152), mirrored training and averaging, NAP $k$, weight decay.

| Arm | bAcc | NLL | Recall G | False G | N→G left | N→G right |
|---|---|---|---|---|---|---|
| d384 mirror k32 wd0.1 | .792 | .577 | .658 | .078 | .127 | .127 |
| d384 mirror k16 wd0.1 | .792 | .577 | .658 | .078 | .127 | .127 |
| d384 mirror k32 wd0.01 | .792 | .577 | .652 | .076 | .125 | .124 |
| d384 mirror k16 wd0.01 | .792 | .577 | .652 | .076 | .125 | .124 |
| d384 mirror k32 wd0.001 | .792 | .578 | .653 | .077 | .124 | .124 |
| d384 mirror k16 wd0.001 | .792 | .578 | .653 | .077 | .124 | .124 |
| d384 mirror k8 wd0.001 | .790 | .578 | .650 | .077 | .126 | .125 |
| d384 mirror k8 wd0.01 | .790 | .579 | .648 | .077 | .126 | .124 |
| d384 mirror k8 wd0.1 | .790 | .578 | .644 | .077 | .125 | .124 |
| d384 plain k32 wd0.01 | .789 | .576 | .768 | .103 | .215 | .141 |
| d1152 mirror k32 wd0.01 | .778 | .603 | .711 | .100 | .176 | .165 |

Table S43: Upgrades of LANTERN's input on development data: dimension (384 or 1152), mirrored training and averaging, NAP $k$, weight decay.

| Arm | bAcc | NLL | Recall G | False G | N→G left | N→G right |
|---|---|---|---|---|---|---|
| d1152 plain k32 wd0.01 | .770 | .603 | .823 | .134 | .296 | .188 |

*Reading.* Mirroring removes the left-right gap and lowers false G but costs recall G; all 1152 dimensions are worse; $k$ and weight decay hardly matter.

### S8.3 Patch-level evidence

LANTERN's head input is the mean of patch tokens, and standardization, projection and the linear head are all linear, so each head's logits split exactly into per-patch contributions: $z_s(x) = \frac{1}{P}\sum_p W_s(I - U_s U_s^\top)x_p + b_s$. Figure S3 maps the glaucoma evidence (G logit minus the larger of N and D, averaged over heads) for the most and least confident glaucoma image of each glaucoma test source. Tokens recomputed for these images matched the stored ones to within 0.13 (feature scale about 90).

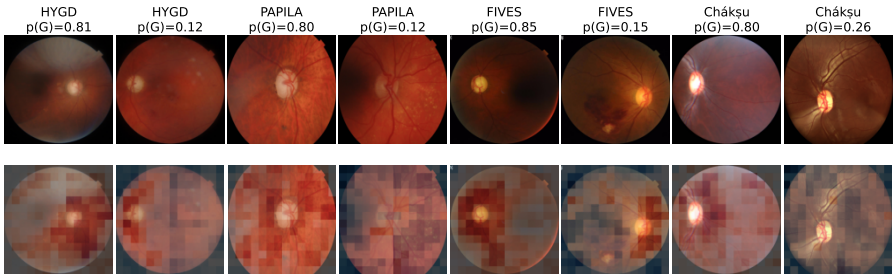



**Fig. S3.** Glaucoma evidence per patch (red: for G, blue: against), for the most and least confident glaucoma image of each source, $p(\mathrm{G})$ from raw fused logits. In confident cases the evidence concentrates on the optic disc; in missed cases it is weak and diffuse. Text scores are global and not mapped.

*Reading.* The maps are exact decompositions of the head, not post-hoc saliency, but they are not a validated localization of pathology.

### S8.4 Later attempts

Two further components were tried with selection rules written before their results were read. *Label scope*: normal labels from DR-only sources were treated as the set {N, G}, with likelihood $p_\mathrm{N} + p_\mathrm{G}$. *Shift-aware reference*: temperature, thresholds and conformal quantiles were fitted on leave-one-source-out predictions rather than on Cal-B, with an optional normal certificate.

**Table S23.** Pairwise AUROC, mean over published test sources holding both classes, with bootstrap intervals over sources.

| Pair | A | B | Sources | AUROC A | AUROC B | A−B | 95% CI |
|---|---|---|---|---|---|---|---|
| D–G | 16-shot probe | Fine-tune | 3 | .939 | .916 | +0.023 | [-0.029, +0.077] |
| D–G | 16-shot probe | Linear probe | 3 | .939 | .917 | +0.022 | [-0.012, +0.059] |
| D–G | 16-shot probe | Group DRO | 3 | .939 | .942 | -0.003 | [-0.035, +0.033] |
| D–G | Linear probe | Fine-tune | 3 | .917 | .916 | +0.001 | [-0.062, +0.081] |
| D–G | Group DRO | ERM | 3 | .942 | .916 | +0.025 | [+0.010, +0.041] |
| N–D | 16-shot probe | Fine-tune | 4 | .946 | .900 | +0.046 | [+0.016, +0.074] |
| N–D | 16-shot probe | Linear probe | 4 | .946 | .928 | +0.019 | [-0.005, +0.053] |
| N–D | 16-shot probe | Group DRO | 4 | .946 | .923 | +0.024 | [+0.007, +0.045] |
| N–D | Linear probe | Fine-tune | 4 | .928 | .900 | +0.028 | [-0.012, +0.069] |
| N–D | Group DRO | ERM | 4 | .923 | .909 | +0.013 | [-0.021, +0.054] |
| N–G | 16-shot probe | Fine-tune | 4 | .749 | .643 | +0.106 | [+0.038, +0.203] |
| N–G | 16-shot probe | Linear probe | 4 | .749 | .678 | +0.071 | [-0.011, +0.152] |
| N–G | 16-shot probe | Group DRO | 4 | .749 | .704 | +0.045 | [-0.006, +0.118] |
| N–G | Linear probe | Fine-tune | 4 | .678 | .643 | +0.035 | [-0.059, +0.135] |
| N–G | Group DRO | ERM | 4 | .704 | .644 | +0.060 | [-0.015, +0.123] |

**Table S24.** Trained on development sources against trained within the target (logistic regression on frozen features, mean over nine encoders; within-source by cross-validation). Gap: within minus development bAcc, with its bootstrap interval.

| | bAcc | | ECE | | AUROC N–G | | AUROC N–D | | |
|---|---|---|---|---|---|---|---|---|---|
| Source | Dev | Within | Dev (TS) | Within | Dev | Within | Dev | Within | Gap |
| AFIO | .516 | .635 | .344 | .153 | .603 | .745 | .856 | .829 | +0.126 [-0.001, +0.249] |
| Chákṣu | .352 | .718 | .261 | .044 | .638 | .806 | – | – | +0.372 [+0.328, +0.415] |
| FIVES | .581 | .857 | .227 | .049 | .780 | .961 | .883 | .969 | +0.283 [+0.245, +0.320] |
| HRF | .583 | .779 | .295 | .182 | .477 | .854 | .926 | .927 | +0.193 [+0.053, +0.328] |
| PAPILA | .394 | .757 | .212 | .070 | .635 | .832 | – | – | +0.371 [+0.296, +0.445] |
| Paraguay | .817 | .928 | .046 | .018 | – | – | .934 | .975 | +0.115 [+0.095, +0.137] |

Table S46: Thresholds from Cal-B against the leave-one-source-out reference, 50 rotations, with and without the normal certificate.

| Fitted on | Target | Test cov. | Test risk | FR | G→N | Chákṣu cov. | Chákṣu risk |
|---|---|---|---|---|---|---|---|
| Cal-B | cov. .80 | .744 | .117 | .076 | .097 | .464 | .444 |
| Cal-B | cov. .90 | .868 | .153 | .108 | .160 | .718 | .463 |
| Cal-B | risk .10 | .641 | .100 | .060 | .085 | .357 | .411 |
| Cal-B + cert. | cov. .80 | .601 | .109 | .035 | .036 | .338 | .601 |
| Cal-B + cert. | cov. .90 | .661 | .145 | .037 | .040 | .475 | .676 |
| Cal-B + cert. | risk .10 | .540 | .094 | .031 | .034 | .263 | .528 |
| LOSO reference | cov. .80 | .830 | .141 | .096 | .139 | .639 | .459 |
| LOSO reference | cov. .90 | .913 | .167 | .120 | .186 | .827 | .469 |
| LOSO reference | risk .10 | .555 | .074 | .040 | .042 | .194 | .402 |
| LOSO reference + cert. | cov. .80 | .650 | .131 | .034 | .035 | .432 | .648 |
| LOSO reference + cert. | cov. .90 | .690 | .158 | .034 | .035 | .529 | .696 |
| LOSO reference + cert. | risk .10 | .524 | .072 | .031 | .031 | .177 | .436 |

*Reading.* Label scope over-calls glaucoma at new sources. The shift-aware reference keeps coverage and risk promises far more often; risk at matched coverage is identical for both, so the arms differ only in where the threshold sits.

**Table S25.** More training sources at a fixed size (300 images per class): test and Chákṣu bAcc for $K = 1, 2, 4, 7$ development sources (mean over random pools).

| | Test | | | | Chákṣu | | | |
|---|---|---|---|---|---|---|---|---|
| Encoder | 1 | 2 | 4 | 7 | 1 | 2 | 4 | 7 |
| clip_b16 | .509 | .478 | .508 | .558 | .527 | .446 | .431 | .494 |
| convnext_t | .481 | .514 | .518 | .557 | .514 | .491 | .408 | .440 |
| dinov2_b14 | .544 | .618 | .593 | .720 | .535 | .555 | .485 | .543 |
| flair[†] | .555 | .633 | .678 | .758 | .511 | .624 | .643 | .679 |
| medsiglip | .594 | .725 | .743 | .817 | .534 | .561 | .504 | .350 |
| resnet50 | .547 | .544 | .503 | .628 | .564 | .552 | .362 | .455 |
| siglip2_l16 | .512 | .518 | .548 | .602 | .494 | .456 | .387 | .102 |
| vit_b16_in21k | .417 | .478 | .542 | .576 | .504 | .456 | .347 | .430 |

**Table S26.** Learning curve: change in test bAcc at $k$ labels per class against the next smaller budget, with bootstrap intervals.

| Encoder | $k$ | ΔbAcc | 95% CI |
|---|---|---|---|
| medsiglip | 16 | -0.008 | [-0.161, +0.130] |
| medsiglip | 64 | +0.021 | [-0.111, +0.124] |
| medsiglip | 128 | +0.040 | [-0.060, +0.120] |
| medsiglip | 512 | +0.009 | [-0.054, +0.078] |
| dinov2_b14 | 16 | -0.130 | [-0.257, +0.032] |
| dinov2_b14 | 64 | -0.044 | [-0.151, +0.088] |
| dinov2_b14 | 128 | -0.016 | [-0.114, +0.089] |
| dinov2_b14 | 512 | +0.003 | [-0.076, +0.060] |
| siglip2_l16 | 16 | -0.050 | [-0.129, +0.024] |
| siglip2_l16 | 64 | -0.015 | [-0.096, +0.087] |
| siglip2_l16 | 128 | +0.025 | [-0.080, +0.125] |
| siglip2_l16 | 512 | +0.007 | [-0.103, +0.056] |

# S9 Statistical Comparisons

Table S47: Paired bootstrap over leakage groups on the published split, 2,000 draws, Holm within each metric; differences are LANTERN minus baseline.

| Metric | Baseline | Baseline value | Diff. | 95% CI | Adj. $p$ |
|---|---|---|---|---|---|
| bAcc (raw) | 16-shot probe, MedSigLIP | .748 | +0.069 | [+0.054, +0.084] | 0.009 |
| bAcc (raw) | Linear probe, MedSigLIP | .728 | +0.089 | [+0.073, +0.103] | 0.009 |
| bAcc (raw) | Zero-shot, MedSigLIP | .709 | +0.108 | [+0.093, +0.124] | 0.009 |
| bAcc (raw) | ERM, DINOv2 LoRA | .707 | +0.109 | [+0.093, +0.125] | 0.009 |
| bAcc (raw) | Group DRO, ConvNeXt-T | .707 | +0.110 | [+0.096, +0.125] | 0.009 |
| bAcc (raw) | Fine-tune (NLL rule), ConvNeXt-T | .677 | +0.140 | [+0.123, +0.156] | 0.009 |
| bAcc (raw) | CORAL, ConvNeXt-T | .676 | +0.140 | [+0.123, +0.157] | 0.009 |
| bAcc (raw) | SWAD, ConvNeXt-T | .672 | +0.145 | [+0.127, +0.161] | 0.009 |
| bAcc (raw) | Fine-tune, Swin-T | .655 | +0.162 | [+0.143, +0.179] | 0.009 |
| bAcc (raw) | WiSE-FT, SigLIP2-L | .634 | +0.182 | [+0.164, +0.200] | 0.009 |
| bAcc (raw) | MLP probe, DINOv2-B | .598 | +0.218 | [+0.198, +0.237] | 0.009 |
| bAcc (raw) | CoOp ctx-init, BiomedCLIP | .497 | +0.320 | [+0.301, +0.338] | 0.009 |
| bAcc (raw) | CoOp + LP, SigLIP-B | .496 | +0.321 | [+0.303, +0.339] | 0.009 |
| bAcc (raw) | Tip-Adapter-F, BiomedCLIP | .495 | +0.322 | [+0.301, +0.343] | 0.009 |
| bAcc (raw) | CoOp, SigLIP-B | .491 | +0.325 | [+0.307, +0.344] | 0.009 |
| bAcc (raw) | Tip-Adapter, BiomedCLIP | .478 | +0.339 | [+0.318, +0.359] | 0.009 |
| ECE (TS) | Zero-shot, MedSigLIP | .032 | -0.012 | [-0.025, +0.005] | 0.221 |
| ECE (TS) | 16-shot probe, MedSigLIP | .044 | -0.024 | [-0.032, -0.009] | 0.009 |
| ECE (TS) | Linear probe, MedSigLIP | .079 | -0.060 | [-0.071, -0.038] | 0.009 |

Table S47: Paired bootstrap over leakage groups on the published split, 2,000 draws, Holm within each metric; differences are LANTERN minus baseline.

| Metric | Baseline | Baseline value | Diff. | 95% CI | Adj. $p$ |
|---|---|---|---|---|---|
| ECE (TS) | Group DRO, ConvNeXt-T | .091 | -0.072 | [-0.084, -0.052] | 0.009 |
| ECE (TS) | Tip-Adapter, BiomedCLIP | .091 | -0.072 | [-0.083, -0.052] | 0.009 |
| ECE (TS) | ERM, DINOv2 LoRA | .115 | -0.096 | [-0.108, -0.075] | 0.009 |
| ECE (TS) | Fine-tune (NLL rule), ConvNeXt-T | .133 | -0.114 | [-0.126, -0.092] | 0.009 |
| ECE (TS) | CORAL, ConvNeXt-T | .136 | -0.116 | [-0.130, -0.094] | 0.009 |
| ECE (TS) | SWAD, ConvNeXt-T | .139 | -0.119 | [-0.133, -0.096] | 0.009 |
| ECE (TS) | Tip-Adapter-F, BiomedCLIP | .147 | -0.127 | [-0.142, -0.108] | 0.009 |
| ECE (TS) | CoOp ctx-init, BiomedCLIP | .150 | -0.131 | [-0.145, -0.107] | 0.009 |
| ECE (TS) | WiSE-FT, SigLIP2-L | .168 | -0.149 | [-0.163, -0.126] | 0.009 |
| ECE (TS) | Fine-tune, Swin-T | .171 | -0.151 | [-0.165, -0.128] | 0.009 |
| ECE (TS) | MLP probe, DINOv2-B | .172 | -0.152 | [-0.166, -0.127] | 0.009 |
| ECE (TS) | CoOp, SigLIP-B | .175 | -0.156 | [-0.171, -0.131] | 0.009 |
| ECE (TS) | CoOp + LP, SigLIP-B | .176 | -0.156 | [-0.172, -0.132] | 0.009 |
| NLL (TS) | 16-shot probe, MedSigLIP | .644 | -0.161 | [-0.186, -0.136] | 0.009 |
| NLL (TS) | Zero-shot, MedSigLIP | .690 | -0.208 | [-0.227, -0.188] | 0.009 |
| NLL (TS) | Linear probe, MedSigLIP | .694 | -0.212 | [-0.248, -0.173] | 0.009 |
| NLL (TS) | Group DRO, ConvNeXt-T | .926 | -0.444 | [-0.511, -0.377] | 0.009 |
| NLL (TS) | ERM, DINOv2 LoRA | .944 | -0.462 | [-0.526, -0.404] | 0.009 |
| NLL (TS) | CORAL, ConvNeXt-T | 1.005 | -0.523 | [-0.589, -0.460] | 0.009 |
| NLL (TS) | Tip-Adapter, BiomedCLIP | 1.020 | -0.537 | [-0.567, -0.506] | 0.009 |
| NLL (TS) | Fine-tune (NLL rule), ConvNeXt-T | 1.036 | -0.553 | [-0.624, -0.485] | 0.009 |
| NLL (TS) | SWAD, ConvNeXt-T | 1.056 | -0.574 | [-0.647, -0.502] | 0.009 |
| NLL (TS) | Tip-Adapter-F, BiomedCLIP | 1.099 | -0.617 | [-0.652, -0.581] | 0.009 |
| NLL (TS) | Fine-tune, Swin-T | 1.174 | -0.691 | [-0.766, -0.619] | 0.009 |
| NLL (TS) | CoOp + LP, SigLIP-B | 1.180 | -0.698 | [-0.752, -0.644] | 0.009 |
| NLL (TS) | CoOp, SigLIP-B | 1.180 | -0.698 | [-0.751, -0.645] | 0.009 |
| NLL (TS) | CoOp ctx-init, BiomedCLIP | 1.189 | -0.707 | [-0.765, -0.649] | 0.009 |
| NLL (TS) | WiSE-FT, SigLIP2-L | 1.243 | -0.761 | [-0.837, -0.684] | 0.009 |
| NLL (TS) | MLP probe, DINOv2-B | 1.259 | -0.777 | [-0.873, -0.681] | 0.009 |

## S10 Reproducibility

Code, saved result tables and this document are archived at `https://doi.org/10.5281/zenodo.23911966` (GitHub: `https://github.com/dangaura-tech/fundusshift`). The README maps each result to its script and table file. The evaluation protocol, including the selection rules written before each later attempt, is `docs/protocol.md`. Images are obtained from their original publishers; split tables are available from the author on request.

**Table S27.** Test-time adaptation per held-out source: test bAcc for none, BatchNorm statistics, TENT and T3A ($K$ = 20, 100); Chákṣu bAcc for none, TENT and T3A-100.

| Setting | Model | None | BN | TENT | T3A-20 | T3A-100 | Ch. none | Ch. TENT | Ch. T3A |
|---|---|---|---|---|---|---|---|---|---|
| Fine-tune | swin_t | .655 | – | .655 | .666 | .668 | .298 | .297 | .309 |
| Fine-tune | resnet50 | .661 | .598 | .598 | .632 | .641 | .417 | .363 | .306 |
| Group DRO | convnext_t | .707 | – | .709 | .705 | .703 | .417 | .430 | .358 |
| ERM | dinov2_lora | .708 | – | .710 | .644 | .662 | .420 | .421 | .298 |
| 16-shot probe | medsiglip | .748 | – | – | .659 | .683 | .675 | – | .496 |
| Linear probe | medsiglip | .728 | – | – | .679 | .679 | .347 | – | .271 |

**Table S28.** Prior correction on top of TS: test bAcc with EM (Saerens et al.) pooled or per source, and with true priors pooled or per source (oracle, uses test labels); ECE for TS, EM per source and oracle per source.

| Setting | Model | TS | EM pooled | EM per src. | True pooled | Oracle per src. | ECE TS | ECE EM | ECE oracle |
|---|---|---|---|---|---|---|---|---|---|
| Zero-shot | medsiglip | .709 | .540 | .635 | .709 | .863 | .032 | .287 | .033 |
| CoOp | siglip_b16 | .491 | .398 | .457 | .513 | .737 | .175 | .433 | .050 |
| CoOp ctx-init | biomedclip | .497 | .416 | .438 | .495 | .754 | .150 | .389 | .040 |
| CoOp + LP | siglip_b16 | .496 | .394 | .450 | .520 | .736 | .176 | .445 | .053 |
| 16-shot probe | medsiglip | .748 | .683 | .787 | .769 | .853 | .044 | .185 | .051 |
| Linear probe | medsiglip | .728 | .778 | .835 | .778 | .853 | .079 | .120 | .081 |
| MLP probe | dinov2_b14 | .598 | .610 | .618 | .621 | .821 | .172 | .226 | .070 |
| Tip-Adapter-F (acc. rule) | biomedclip | .495 | .407 | .434 | .511 | .739 | .147 | .389 | .047 |
| Tip-Adapter (acc. rule) | biomedclip | .478 | .333 | .495 | .460 | .779 | .091 | .292 | .120 |
| Tip-Adapter-F | biomedclip | .495 | .407 | .434 | .511 | .739 | .147 | .389 | .047 |
| Tip-Adapter | biomedclip | .478 | .333 | .495 | .460 | .779 | .091 | .292 | .120 |
| Fine-tune | swin_t | .655 | .657 | .661 | .675 | .827 | .171 | .226 | .099 |
| Fine-tune, NLL rule | convnext_t | .677 | .687 | .710 | .707 | .842 | .133 | .182 | .073 |
| CORAL | convnext_t | .676 | .687 | .704 | .704 | .851 | .136 | .172 | .069 |
| SWAD | convnext_t | .672 | .691 | .704 | .706 | .851 | .139 | .166 | .060 |
| Group DRO | convnext_t | .707 | .720 | .723 | .733 | .857 | .091 | .188 | .051 |
| ERM | dinov2_lora | .707 | .655 | .726 | .727 | .826 | .115 | .218 | .104 |
| WiSE-FT | siglip2_l16 | .634 | .598 | .627 | .669 | .813 | .168 | .281 | .103 |

**Table S30.** CLAHE against unprocessed images, same seeds and hardware.

| Setting | Config | Preprocessing | Val | Test | Eq-src | Chákṣu | ECE TS | Rec. G |
|---|---|---|---|---|---|---|---|---|
| Fine-tune | convnext_t_ce | none | .886 | .668 | .612 | .273 | .133 | .385 |
| Fine-tune | convnext_t_ce | CLAHE | .876 | .653 | .604 | .234 | .132 | .293 |
| Fine-tune | swin_t_mbls | none | .893 | .655 | .600 | .298 | .171 | .411 |
| Fine-tune | swin_t_mbls | CLAHE | .882 | .650 | .608 | .283 | .152 | .351 |
| Linear probe | medsiglip_class_ls | none | .920 | .728 | .654 | .347 | .079 | .488 |
| Linear probe | medsiglip_class_ls | CLAHE | .911 | .788 | .702 | .400 | .041 | .663 |
| 16-shot probe | medsiglip_none_ce | none | .802 | .748 | .683 | .675 | .044 | .783 |
| 16-shot probe | medsiglip_none_ce | CLAHE | .790 | .747 | .693 | .678 | .033 | .804 |

**Table S31.** Referral rules at a threshold for 80% coverage on Cal-B: maximum softmax (MSP), KNN distance in feature space, or both. Open: share of the 861 out-of-scope images accepted.

| Setting | Model | Score | Test cov. | Test risk | Cal-B cov. | Open | G→N |
|---|---|---|---|---|---|---|---|
| Fine-tune | swin_t | MSP | .694 | .250 | .801 | .666 | .147 |
| Fine-tune | swin_t | KNN | .415 | .084 | .801 | .165 | .007 |
| Fine-tune | swin_t | MSP+KNN | .484 | .113 | .802 | .252 | .020 |
| Group DRO | convnext_t | MSP | .738 | .182 | .801 | .602 | .089 |
| Group DRO | convnext_t | KNN | .400 | .076 | .801 | .101 | .010 |
| Group DRO | convnext_t | MSP+KNN | .487 | .094 | .802 | .187 | .026 |
| ERM | dinov2_lora | MSP | .715 | .223 | .801 | .610 | .067 |
| ERM | dinov2_lora | KNN | .549 | .266 | .801 | .216 | .113 |
| ERM | dinov2_lora | MSP+KNN | .628 | .221 | .804 | .290 | .092 |
| Linear probe | medsiglip | MSP | .610 | .129 | .801 | .337 | .061 |
| Linear probe | medsiglip | KNN | .419 | .227 | .801 | .059 | .093 |
| Linear probe | medsiglip | MSP+KNN | .421 | .153 | .802 | .074 | .057 |
| 16-shot probe | medsiglip | MSP | .753 | .195 | .801 | .679 | .012 |
| 16-shot probe | medsiglip | KNN | .419 | .223 | .801 | .059 | .027 |
| 16-shot probe | medsiglip | MSP+KNN | .483 | .183 | .802 | .105 | .020 |

**Table S32.** Out-of-scope detection AUROC from penultimate features of trained models (SWAD rows omitted: its saved checkpoint holds pre-averaging weights).

| Setting | Model | MSP | MaxLogit | Energy | KNN | Mahalanobis |
|---|---|---|---|---|---|---|
| Fine-tune | swin_t | .549 | .567 | .572 | .664 | .754 |
| Group DRO | convnext_t | .618 | .610 | .606 | .745 | .791 |
| ERM | dinov2_lora | .613 | .610 | .610 | .753 | .754 |

**Table S33.** Multiple-choice answering by multimodal LLMs (mean over six option orders).

| Model | Test bAcc | Rec. N | Rec. D | Rec. G | Chákṣu bAcc | Chákṣu Rec. G |
|---|---|---|---|---|---|---|
| Fundus-R1-7B | .576 | .789 | .795 | .143 | .463 | .128 |
| Qwen2.5-VL-7B-Instruct | .424 | .926 | .347 | .000 | .488 | .005 |
| llava-med-v1.5-mistral-7b-hf | .347 | .905 | .056 | .080 | .507 | .037 |
| llava-onevision-qwen2-7b-ov-hf | .407 | .367 | .855 | .000 | .096 | .000 |

**Table S34.** Recall of D by DR grade on test (grade −1: Davis grading without a numeric ICDR grade).

| Setting | Model | −1 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|---|
| Zero-shot | medsiglip | .785 | .000 | .734 | .954 | .838 |
| CoOp | siglip_b16 | .711 | .444 | .688 | .823 | .915 |
| CoOp ctx-init | biomedclip | .808 | .111 | .618 | .884 | .925 |
| CoOp + LP | siglip_b16 | .718 | .444 | .700 | .835 | .929 |
| 16-shot probe | medsiglip | .767 | .056 | .841 | .989 | .967 |
| Linear probe | medsiglip | .727 | .000 | .844 | .981 | .977 |
| MLP probe | dinov2_b14 | .665 | .222 | .737 | .935 | .932 |
| Tip-Adapter-F (acc. rule) | biomedclip | .660 | .167 | .569 | .831 | .885 |
| Tip-Adapter (acc. rule) | biomedclip | .368 | .000 | .122 | .432 | .499 |
| Tip-Adapter-F | biomedclip | .660 | .167 | .569 | .831 | .885 |
| Tip-Adapter | biomedclip | .368 | .000 | .122 | .432 | .499 |
| Fine-tune | swin_t | .724 | .389 | .844 | .974 | .986 |
| Fine-tune, NLL rule | convnext_t | .717 | .389 | .801 | .965 | .969 |
| CORAL | convnext_t | .704 | .278 | .801 | .966 | .972 |
| SWAD | convnext_t | .724 | .278 | .817 | .966 | .972 |
| Group DRO | convnext_t | .712 | .111 | .743 | .969 | .952 |
| ERM | dinov2_lora | .701 | .333 | .865 | .980 | .996 |
| WiSE-FT | siglip2_l16 | .695 | .222 | .755 | .964 | .968 |

**Table S35.** Glaucoma recall on clean test images and at the strongest blur and JPEG levels.

| Setting | Model | Clean | Blur | JPEG |
|---|---|---|---|---|
| Fine-tune | swin_t | .419 | .071 | .100 |
| Group DRO | convnext_t | .447 | .026 | .073 |
| ERM | dinov2_lora | .598 | .438 | .358 |
| 16-shot probe | medsiglip | .784 | .267 | .870 |
| Linear probe | medsiglip | .491 | .144 | .086 |

**Table S36.** Complete specification of LANTERN.

| | |
|---|---|
| Encoder | MedSigLIP (patch 14, 448 px, 27 layers, width 1152), frozen |
| Head input | Last hidden state, $32 \times 32$ tokens pooled to $14 \times 14$; fixed random orthonormal map to 384 dimensions (seed 0); grid mean |
| Standardization | Per dimension, training-split mean and standard deviation |
| Projection | Within-class between-site means, at most 32 directions, fitted per head on its training rows |
| Heads | One linear head per development source, trained without it; 1,155 parameters each |
| Training | Label smoothing 0.1; class-balanced sampling; AdamW, lr $3 \times 10^{-4}$, wd $10^{-2}$; batch 64; 12 epochs; best epoch by validation bAcc on the head's own sites |
| Rescaling | Centre per image; divide by the standard deviation over validation rows |
| Text | Mean of the three banks of Table S5, on the pooled image embedding |
| Fusion | 0.5 heads + 0.5 text; TS on Cal-A |
| Seeds | 0, 1, 2 on the published split; one per rotation |

**Table S39.** LANTERN on each held-out development site.

| Held-out site | bAcc | Recall G | False G |
|---|---|---|---|
| Bangladesh Eye | .751 | .716 | .149 |
| DeepDRiD | .776 | – | .081 |
| GRAPE | .812 | .812 | – |
| IDRiD | .758 | – | .026 |
| Jichi | .767 | – | .101 |
| OIA-DDR | .730 | – | .202 |
| SUSTech-SYSU | .946 | – | .040 |

**Table S40.** Designs tried and not retained (development). Reference: pooled-feature pipeline 0.791; single cross-entropy head 0.675.

| Experiment | bAcc | Reading |
|---|---|---|
| Batch network predicting temperature and class bias (1 seed) | .539 | Below the raw head (.666) |
| EM prior shift with TS (1 seed) | .523 | Unreliable with few classes per source |
| Polar Fourier features around the disc | .702 | Below the pair loss alone (.723) |
| Site prior, $\kappa$ from 30 to 3000 | .693–.717 | No gain |
| Full residual corrector over logits | .667 | Recall G fell to 0 |
| Corrector with fusion weights only | .781 | ECE .057, bAcc did not rise |
| RPAttn head on three token depths (1 seed) | .664 | Recall G .008; with text .690 |
| RPAttn + MixStyle (1 seed) | .614 | Recall G .012; with text .688 |
| N/D from RPAttn, G from low-capacity branch (1 seed) | .788 | .793 for the pooled pipeline on that seed |
| Late fusion of nine encoder inputs | .765 | Other encoders lowered the score |
| Greedy nested encoder selection | .789 | Mostly kept MedSigLIP alone |
| Magnified optic-disc view with whole image | .768 | Recall G up (.767), false G up (.135) |
| Image-conditioned fusion gate | .763 | Below a plain average |
| Global class offset chosen on other sites | .679 | GRAPE fell to .006 |
| ConvNeXt-T end-to-end with pair loss (1 seed) | .622 | GRAPE .004 |

**Table S41.** Held-out development sites for the pooled-feature pipeline: head, text and fusion bAcc, recalls, and images wrong under all nine encoder inputs.

| Held-out site | *n* | Head | Text | Fusion | N | D | G | All wrong |
|---|---|---|---|---|---|---|---|---|
| Bangladesh Eye | 3,153 | .651 | .732 | .753 | .761 | .963 | .536 | .039 |
| OIA-DDR | 11,269 | .660 | .738 | .735 | .685 | .786 | – | .137 |
| DeepDRiD | 1,804 | .803 | .721 | .795 | .682 | .908 | – | .111 |
| GRAPE | 528 | .644 | .750 | .741 | – | – | .741 | .104 |
| IDRiD | 462 | .757 | .752 | .751 | .527 | .974 | – | .100 |
| Jichi | 8,475 | .779 | .736 | .803 | .824 | .781 | – | .103 |
| SUSTech-SYSU | 1,097 | .975 | .889 | .957 | .938 | .975 | – | .009 |

**Table S42.** Normal images predicted G, by disc side (automatic, label-free disc location), held-out site.

| Held-out site | Disc left | Disc right |
|---|---|---|
| OIA-DDR | .386 | .191 |
| Bangladesh Eye | .141 | .023 |
| IDRiD | .119 | .045 |
| SUSTech-SYSU | .086 | .019 |
| Jichi | .067 | .054 |
| DeepDRiD | .096 | .086 |

**Table S44.** Post-processing checked on held-out development sites, fitted on calibration rows of the other sites.

| State | ECE | NLL | Cov 80 cov. | Cov 80 risk | Risk 10 risk | LAC cov. | LAC size | Class-LAC | APS cov./size |
|---|---|---|---|---|---|---|---|---|---|
| Raw | .148 | .576 | .720 | .105 | .094 | .866 | 1.18 | .846 | .930/1.62 |
| TS | .052 | .502 | .719 | .104 | .096 | .867 | 1.18 | .842 | .946/1.78 |
| FTS | .053 | .505 | .720 | .106 | .096 | .867 | 1.18 | .843 | .919/1.61 |

**Table S45.** Label scope: development (scope-aware bAcc counts a DR-only normal predicted G as correct) and the 50 rotations.

| Arm | bAcc | Scope-aware bAcc | ECE | Recall G | N→G, DR-only | N→G, full scope |
|---|---|---|---|---|---|---|
| exact | .792 | .844 | .151 | .764 | .145 | .322 |
| scope | .776 | .864 | .136 | .811 | .244 | .255 |
| exact (rotations) | .771 | .801 | .033 | .667 | – | .248 |
| scope (rotations) | .761 | .831 | .040 | .797 | – | .443 |

**Table S48.** Configurations with development bAcc of at least 0.791, all evaluated on the published split.

| Configuration | Dev | Test bAcc | Eq-src | Recall G | NLL (TS) | Chákṣu |
|---|---|---|---|---|---|---|
| Mirror, weight decay $10^{-3}$ | .800 | .809 | .726 | .696 | .503 | .714 |
| Mirror, weight decay $10^{-2}$ | .799 | .808 | .727 | .708 | .502 | .726 |
| Mirror, weight decay $10^{-4}$ | .795 | .807 | .724 | .681 | .511 | .703 |
| Cross-site pair loss | .794 | .809 | .717 | .722 | .507 | .694 |
| Same-site pair loss | .792 | .791 | .696 | .670 | .524 | .713 |
| **Tokens, weight decay $10^{-2}$ (LANTERN)** | .792 | .817 | .724 | .765 | .482 | .731 |
| Pooled feature, weight decay $10^{-4}$ | .791 | .799 | .706 | .687 | .520 | .684 |
| Pooled feature + tokens | .791 | .812 | .719 | .719 | .493 | .701 |
| Disc side canonicalised | .791 | .805 | .733 | .679 | .511 | .637 |
| Mirrored text only | .791 | .800 | .714 | .679 | .512 | .704 |